A GUIDE TO PRODUCT QUALITY STANDARDS

# WOOD PRODUCTS

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## DISCLAIMER



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Also, the references to any specific process(es) and organisation(s) or institution(s) in the compendium do not constitute their endorsement or recommendation, in any way.

## ABBREVATIONS

| | |
|---|---|
| *ACCC* | Australian Competition and Consumer Commission |
| *ANSI* | American National Standards Institute |
| *APLAC* | Asia Pacific Laboratory Accreditation Co-operation |
| *APHIS* | Animal and Plant Health Inspection Service |
| *AS* | Australian Standards |
| *ASTM* | American Society for Testing and Materials |
| *BICON* | Bio Security Import Condition System |
| *BIS* | Bureau of Indian Standards |
| *CE* | Conformité Européenne |
| *CoC* | Certificate of Conformity |
| *CPSC* | Consumer Products Safety Commission |
| *CPSIA* | Consumer Product Safety Improvement Act |
| *CSPA* | Consumer Product Safety Act |
| *DOC* | Declaration of Compliance |
| *EC* | Council Regulation (of the European Parliament and Council) |
| *ECHA* | European Chemicals Agency |
| *EN* | European Standards |
| *EPA* | Environment Protection Act |
| *EU* | European Union |
| *EUTR* | European Timber Regulation |
| *FCM* | Food Contact Materials |
| *FDA* | Food and Drug Administration |
| *FSHA* | Federal Hazardous Substances Act |
| *FSSAI* | Food Safety and Standards Authority of India |
| *GCC* | Gulf Co-operation Council |
| *GI* | Geographical Indication |
| *GSO* | GCC Standardization Organization |
| *GPSD* | General Product Safety Directive |
| *IS* | Indian Standard |
| *ISI* | Indian Standards Institution |
| *ILAC* | International Laboratory Accreditation Co-operation |
| *ISO* | International Organization for Standardization |
| *MDF* | Medium Density Fibre |
| *MLA* | Multilateral Recognition Arrangement |
| *MRA* | Mutual Recognition Arrangement |
| *NABCB* | National Accreditation Board for Certification Bodies |
| *NABL* | National Accreditation Board for Testing and Calibration Laboratories |
| *REACH* | Registration, Evaluation, Authorization and Restriction of Chemicals |
| *RoHS* | Restriction of Hazardous Substances Directive |
| *SASO* | Saudi Standards, Metrology and Quality Organization |
| *SRL* | Specific Release Limits |
| *QCI* | Quality Council of India |
| *QCO* | Quality Control Order |

## BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This compendium brings relevant regulations and standards pertaining to the wooden crafts and goods. Uttar Pradesh is famed for its wood crafts which are created with different types of woods which include household items like pots, trays, bowls and decorative pieces with intricate etching.

# WOODEN BED, BASTI, MAHARAJ GANJ, SHRAVASTI, SAHARANPUR

## 1. INTRODUCTION

The bed is considered the most important piece of furniture in the household. It provides the key place for relaxing.

Various districts of Uttar Pradesh are known for furniture works. Basti, Saharanpur and Maharaj Ganj districts have sufficient availability of raw materials for wood craft and furniture making and are major furniture manufacturing centres. These items are made by skilled craftsmen are sought after locally as well as across the country. Yet another district in Uttar Pradesh i.e., Shravasti district is well famed for furniture manufacturing.

The beds are usually made of hard-wood to get a better durability and smooth finishing. Types of woods used in India are: Teakwood, Salwood, Rosewood, Marandi and many more.

Most hardwoods used for woodwork that are kiln-dried (A wood that is dried and is free from moisture) are acceptable as a measure of quality for the bed. Some popular and reliable hardwoods are oak and maple, due to their ability to age well with strong durability. Teak wood is considered as one of the best materials which is a hard wood and provides good longevity.

## 2. RAW MATERIALS

The raw materials used for making beds are wood, timber, plywood, veneer, glue and paints, polishes, screws, nails and assorted hardware's as per design.

## 3. MANUFACTURING PROCESS

Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in use. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength. The manufacturing process for beds is as under:

### 3.1. Selection of wood

Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen.

### 3.2. Cutting of wood

The chosen wood is cut in specific dimensions. These sizes of beds are globally recognized as King Size and Queen Sized beds, though custom lengths are also made. Contemporary beds usually have the frame built in solid wood (comprising of the legs and the other components of the frame), the rest is made using plywood/MDF. This is more an economic measure, keeping in mind the costs of the wooden materials. Beds can also be designed to have storage compartments built into them and such beds are called 'box beds'.

### 3.3. Carving

The designs are traced on to the wooden members and using gouges, chisels and other wood working tools, the excess wood is shaved away. The final design is sandpapered and smoothened.

### 3.4. Frame building

After the designs have been carved on the individual structural members, the frame is built using the key structural members and assembled using various joinery details. Nails and bolts are used to fasten the key structural members rigidly together or add a decorative headrest which can be attached to the frame of the bed. Alternatively, furniture brackets can be utilized to give a joint free exterior finish.

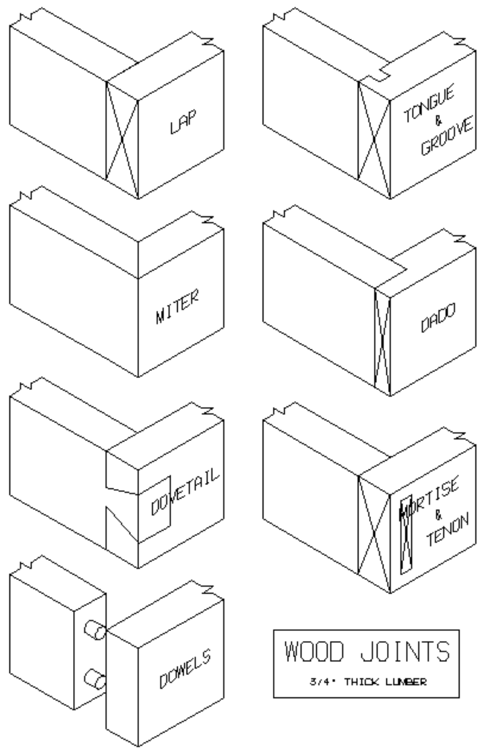


**Figure 1:** Wooden joinery details

### 3.5. Finishing

Finally, the bed is checked to remove dust, saw dust and unwanted particles. The bed is cleaned and polished as required.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Polish and varnish applied should be uniform, devoid of patches
- The furniture should be free from cracks.
- The joints should be tight.

- The bed should be well balanced with no wobbling.
- The wood should be dry and free from any kind of termites or borers.
- The edges should be rounded.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>b. European Union Timber Regulation (EUTR),2010<br>c. General Product Safety Directive (GPSD) | a. European Commission<br>b. European Chemical Agency (ECHA) | a. Legality of timber<br>b. Traceability<br>c. Procedural framework<br>d. Marking and labelling |
| 2 | United States of America | a. The Lacey Act, 2008<br>b. California Proposition 65, Act of 1986 | a. Consumer Product Safety Commission (CPSC) | a. Nature of Timber (endangered or not)<br>b. The genus and species of tree |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | c. Bureau of Consumer Protection Regulations<br>d. Environment Protection Act (EPA)<br>e. Consumer Product Safety Improvement Act (CPSIA),2008 | b. Animal and Plant Health Inspection Service (APHIS)<br>c. United States Environment Protection Agency (EPA)<br>d. California State Government<br>e. Federal Trade Commission | c. Quantity of regulated material<br>d. Country of origin<br>e. Formaldehyde in wood<br>f. Labelling<br>g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act ,2010<br>c. Illegal Logging Prohibition Act, 2012 | a. Australian Competition and Consumer Commission (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Gulf Technical Regulations: GSO: 450 -1994 | a. GCC standardization organization | a. Definitions of primer paints (based coating) and its types and includes the general regulation and physical properties which shall be available in each type. |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Glossary of terms relating to paints | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood work | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **ISO 24294: 2021** | Vocabulary for timber |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | - | - | Middle East | **GSO 450: 1994** | Specifications for Paints and varnishes |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes –Specification | USA | **ASTM - D4761** | Test method for wood based material |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | European Union | **ISO/DIS 9665** | Specifications for adhesives — Animal glues |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Animal glue for general wood-Working purposes -Specification | USA | **ASTM 3618: 2005: R2015** | Detection of lead in Paints |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 12406: 2021** | Medium density fibre boards for general purpose | USA | **ASTM D143** | Test methods for timber |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture and cabinets-Classification | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | Australia | **AS 2796** | Methods for seasoning of timber |
| | **-** | **-** | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of working quality of timber under different wood working operations- Method of test | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Code of practice | - | - | - |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | European Union | **BS EN 1725: 1998** | Beds: Safety requirements and test methods |
| | **IS 17635: 2022** | Beds, Specification | European Union | **ISO 19833: 2018** | Wooden beds: Stability analysis |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | European Union | **ISO 9098 - 1: 1994** | Bunk beds for domestic use, Part1: Safety Requirements |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | USA | **ASTM F1427 - 21** | Safety requirement for beds |
| | - | - | Australia | **AS 4220** | Wooden beds |
| | - | - | Middle East | **GSO ISO 19833: 2018** | Wooden Beds: Stability analysis |
| | - | - | Middle East | **GSO ISO 1725: 2009** | Beds: Safety requirements and test methods |
| | - | - | European Union | **ISO 4211 -2: 2013** | Test for surface finishes on wooden surfaces |
| | - | - | European Union | **ISO 7170: 2021** | Furniture — Storage units — Determination of strength and durability |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| **General** | **IS 707: 2007 (Reaffirmed Year: 2021)** -Timber technology and utilization of wood, bamboo and cane - Glossary of Terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ Botanical features and other purely scientific terms are excluded from the scope of this standard. |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Code of practice for fire safety of industrial buildings - Saw mills and wood works | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paints and varnishes- Terms and definitions | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Gulf technical regulations |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Glossary of terms in timber-related standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Prescribes requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paint |
| | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes - Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of600-900 kg/m$^3$.<br>▪ This standard does not cover veneered or laminated or pre-laminated or other specially treated boards, moulded boards, etc. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture and cabinets- Classification | ▪ This standard covers the general classification of Indian Timber species suitable for furniture and cabinets. It also lays down the general requirements of quality, moisture content and preservative treatment for the timber. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ Adjunct standards and referenced documents |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ Exclusions contained. |
| | **ISO/DIS 9665** - Adhesives — Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for Plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lbs.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of timber, Part1**:** Standard methods of testing small clear specimens of timber |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **ASTM 3618: 2005: R2015** - Paint**:** Detection of lead | ▪ Detection of lead in paints |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning<br>b. Sanding<br>c. Turning<br>d. Shaping<br>e. Mortising<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products. |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 17635: 2022** - Wooden beds | ▪ This standard lays down requirements of material, functional dimensions and finishes of wooden bed used as domestic furniture. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture**:** Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture**:** Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture**:** Part 3 Resistance to marking by oils and fats<br>▪ Methods of test for clear finishes for wooden furniture**:** Part 4 Resistance to marking by liquids<br>▪ Methods of test for clear finishes for wooden furniture**:** Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI**:** Resistance to Mechanical Damage |

### 6.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **BS EN 1725: 1998** - Domestic furniture - Beds and mattresses – Safety Requirements and Test Methods | ▪ Specifies mechanical safety requirements and testing for all types of fully erected domestic adult beds including all component elements such as bed frame, bed base mattress and mattress pads |
| | **ISO 19833:2018** - Beds- test methods for determining the stability, strength and durability | ▪ Specifies test methods for determining the stability, strength and durability of all types of fully assembled beds including bed frames and bed bases. |
| | **ISO 9098 - 1: 1994** - Bunk beds for domestic use | ▪ Safety requirements and test methods of bunk beds |
| | **ASTM F1427 – 21** - Bunk beds safety | ▪ Safety requirement and specification for bunk beds |
| | **AS 4220** - Beds | ▪ Specification and constructional requirements of bunk beds |
| | **GSO ISO 19833: 2018** - Beds- test methods for determining the stability, strength and durability | ▪ Specifies test methods for determining the stability, strength and durability of all types of fully assembled beds including bed frames and bed bases. |
| | **GSO ISO 1725: 2009** - Domestic furniture - Beds and mattresses – Safety requirements and test methods | ▪ Specifies mechanical safety requirements and testing for all types of fully erected domestic adult beds including all component elements such as bed frame, bed base mattress and mattress pads |
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


1. Product information- http://odopup.in/en/article/saharanpur
2. Product information- http://odopup.in/en/article/basti
3. Product information- http://odopup.in/en/article/shravasti
4. Product information- http://odopup.in/en/article/maharajganj
5. International Legislation/Regulation:
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    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber
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8. EU: https://www.en-standard.eu/
9. USA: https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
10. AUS: https://www.standards.org.au/
11. GULF: https://www.gso.org.sa/store/
12. NABL accredited laboratories: https://nabl-india.org/ , https://parakh.ncog.gov.in
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## CHAIRS, BASTI, RAE BARELI, SAHRANPUR, PILIBHIT

### 1. INTRODUCTION

The chair has been used as an essential article of furniture since antiquity, although for many centuries it was a symbolic article of state and dignity rather than an article for ordinary use. Chairs are amongst the most basic piece of furniture in the Indian household. It serves to be a major furniture for sitting and relaxing. Various districts of Uttar Pradesh are known for furniture works. Saharanpur, Pilibhit, Basti and Rae Bareli are famed for wooden chairs of different kinds.

A chair is a basically a type of seat and its primary features are two pieces of a durable material, attached as back and seat to one another at a 90° or slightly greater angle as per the design, with usually the four corners of the horizontal seat attached in turn to four legs—or other parts of the seat's underside attached to three legs or to a shaft about which a four-arm turnstile on rollers can turn—strong enough to support the weight of a person who sits on the seat (usually wide and broad enough to hold the lower body from the buttocks almost to the knees) and leans against the vertical back (usually high and wide enough to support the back to the shoulder blades). The legs are typically high enough for the seated person's thighs and knees to form a 90° or lesser angle. Used in a number of rooms in homes, in schools and offices (with desks), and in various other workplaces, chairs may be made of wood, metal, or synthetic materials, and either the seat alone or the entire chair may be padded or upholstered in various colours and fabrics. Various types of chairs include:

- Wooden Rocking chair
- Wooden Arm chair
- Wooden Folding chair
- Wooden Carved chair
- Wooden Office work chair
- Wooden Dining chair

The wood crafts of Saharanpur are renowned the world over and have been given a GI tag, which further enhances their presence in the global markets as a unique product.

### 2. RAW MATERIALS

To make a wooden chair, the raw materials used are wooden laminates, timber, glue and paints, polishes, veneer etc.

## 3. MANUFACTURING PROCESS

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.

The basic steps in manufacturing of chairs are mentioned below:

### 3.1. Selection of wood

Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen.

### 3.2. Cutting of wood

The chosen wood is cut in specific dimensions. The sizes of chairs are not standardized and different kinds of chairs, in different sizes are available. The basic anthropometric data governs their minimum sizes and the age group. The wood must stay in a suitable temperature and humidity-controlled room or the wood may swell (too much humidity) or shrink (very dry) and the piece will have cracks when finished. Temperatures must stay in the range of 50-85°F (10-29°C).

### 3.3. Shaping

Using a shaving horse, the chair parts are planed and using draw knives and spoke shaves different designs are engraved on the surface of the wood. Most commonly the chair back or legs are chosen for carving designs.

### 3.4. Steaming and Bending

Steamers are used in bending the back legs and slats. The back legs and slats are now steamed and bent into the shapes that fit the chair type.

### 3.5. Assembly

All the parts joints are glued and fixed with screws. Various joinery details are utilized to assemble the chairs such as tenon and mortise, dovetail joints etc.

### 3.6. Finishing

Finally, the chair is checked to remove dust, saw dust and unwanted particles. The cleaned chair is then polished, cushioned (as required). Veneer may be applied as per the demand and customer requirement. Usually chairs are also tested with the application of a dead load to check for its strength and stability.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Polish and varnish applied should be uniform, devoid of patches
- The furniture should be free from cracks.
- The joints should be tight.
- The wood should be dry and free from any kind of termites or borers.
- The edges should not be sharp.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | **a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulation<br>**b.** European Union Timber Regulation (EUTR) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Legality of timber<br>**b.** Traceability<br>**c.** Procedural framework<br>**d.** Marking and labelling |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | c. General Product Safety Directive (GPSD) | | |
| 2 | United States of America | a. The Lacey Act, 2008<br>b. California Proposition 65, Act of 1986<br>c. Environment Protection Act (EPA)<br>d. Consumer Product Safety Improvement Act (CPSIA), 2008 | a. Consumer Product Safety Commission (CPSC)<br>b. California State Government<br>c. United States Environment Protection Agency (EPA)<br>d. Animal and Plant Health Inspection Service (APHIS ) | a. Nature of Timber (endangered or not)<br>b. The genus and species of tree<br>c. Quantity of regulated material<br>d. Country of origin<br>e. Formaldehyde in wood<br>f. Labelling<br>g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act, 2010<br>c. Illegal Logging Prohibition Act 2012 | a. Australian Competition and Consumer Commission Act (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Product Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Glossary of terms relating to paints | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of Terms | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood work | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **GSO 450: 1994** | Paints and varnishes |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| Raw Materials (Refer Table 2.a & 2.b) | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | European Union | **ISO/DIS 9665** | Animal glue |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** | Converted timber (coniferous) : Part 1 Light furniture | USA | **ASTM - D4761** | Test method for wood based material |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of working quality of timber under different wood working operations- Method of test | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Code of Practice | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture's and cabinets- Classification | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lingnocellulosic materials (medium density) for general purposes -Specification | USA | **ASTM D143** | Test methods for timber |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards: Specification | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Seasoning of timber: code of practice | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| | - | - | Australia | **AS 2796** | Methods for seasoning of timber |
| Manufacturing/ Processing (Refer Table 3.a & 3.b) | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Clear finishes for wooden furniture | European Union | **BS EN 1725: 1998** | Specification: Domestic furniture |
| | **IS 12406: 2021** | Medium density fibre boards | USA | **ASTM F782 - 19** | Specification: Furniture |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes - Recommendations | - | - | - |
| | **IS 3845: 1966 (Reaffirmed Year: 2020)** | Code of practice for joints used in wooden furniture | - | - | - |
| Finishing (Refer Table 4.a & 4.b) | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | European Union | **ISO 9221 - 1: 2015** | Wooden children chairs |
| | **IS 17632: 2022** | Wooden folding chairs | United States of America | **ANSI 7173** | Analysis of strength and durability of chairs |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 5416: Part 1: 1988 (Reaffirmed Year: 2017)** | Methods of test of strength and stability of chairs and stools | USA | **ANSI 7174 - 1: 1988** | Wooden chairs: Determination of stability |
| | - | - | Australia | **AS 4688.1: 2018** | Wooden chairs : Determination of stability |
| | - | - | Middle East | **GSO ISO 9221 - 1: 2021** | Children chairs |
| | - | - | European Union | **ISO 4211 -2: 2013** | Test for surface finishes on wooden surfaces |
| | - | - | European Union | **ISO 7170: 2021** | Wooden furniture |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ Botanical features and other purely scientific terms are excluded from the scope of this standard. |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practice for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paints and varnishes – Terms and definitions | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |

| Stage | Code | Scope |
|---|---|---|
| | **AS 4491:1997** - Glossary of terms related to Timber | ▪ Glossary of terms in timber-related standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Requirements of high density wood particle boards in flat sheet or moulded forms. |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** - Converted timber (coniferous) : Part 1 Light furniture | ▪ Specification and analysis of timbers suitable for furniture making. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning,<br>b. Sanding,<br>c. Turning,<br>d. Shaping,<br>e. Mortising, and<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products. |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered Particle Boards -Specification | ▪ Requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315: 1974, IS 287: 1993 (Reaffirmed Year: 2017) and IS 11215 :1991 respectively. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture's and cabinets-Classification | ▪ Requirement of timber used in furniture making |
| | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Veneered particle boards -Specification | ▪ Requirements such as grades and types, material, manufacture, dimensions and tests for non-veneered particle boards. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/DIS 9665** - Animal glue | ▪ Specification of glues used in wooden furniture's |
| | **GSO ISO 1096: 2021** - Specification for Plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lb.) and the basic SI unit of mass (Kg) are cited. |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **GSO 1424: 2002** -Methods of testing timber | ▪ Methods of testing of timber part1**:** standard methods of testing small clear specimens of timber |

### 6.4. Table 3.a: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ Requirement and specification of medium density fibre board used. |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 3 Resistance to marking by oil and fats<br>▪ Methods of test for clear finishes for wooden furniture: Part 4 Resistance to marking by liquids<br>▪ Methods of test for clear finishes for wooden furniture: Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI: Resistance to Mechanical Damage |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum |

| Stage | Code | Scope |
|---|---|---|
| | moisture content for timber used for different purposes - Recommendations | moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 3845: 1966 (Reaffirmed Year: 2020):** Code of practice for joints used in wooden furniture | ▪ Confirmation and practices about the joinery work |

### 6.5. Table 3.b: (International Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **EN 1725: 2018** - Domestic furniture | ▪ This standard covers the requirements and test for soil based blocks for use in general building construction. |
| | **ASTM F782 – 19** - Furniture | ▪ This specification covers the construction of furniture doors for use where other marine furniture specifications are applicable<br>▪ Specification applies to all furniture doors for marine furniture, in items requiring hinged doors.<br>▪ The values stated in inch-pound units are to be regarded as standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard.<br>▪ Standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee. |

### 6.6. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 17632: 2022** - Wooden folding chairs | ▪ Requirements for materials, sizes, construction and finish of wooden folding chairs. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, operations and workmanship | ▪ Various practises for finishing of wood-based materials |
| | **IS 5416: Part 1: 1988 (Reaffirmed Year: 2017)** - Methods of test for strength and stability of chairs and stools, Part 1: Strength, stability) | ▪ Methods to check and test strength, stability of chairs |

### 6.7. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 9221 - 1: 2015** - Children chairs | ▪ Safety requirements for children chairs |
| | **GSO ISO 9221 - 1: 2021** - Children chairs | ▪ Safety requirements for children chairs |
| | **ANSI 7173** - Chairs | ▪ Strength analysis<br>▪ Durability analysis |
| | **ANSI 7174 - 1: 1988** - Chairs | ▪ Stability analysis |
| | **AS 4688.1: 2018** - Chairs | ▪ Stability analysis |
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Part of ISO 4211 specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

## CUPBOARDS, SHRAVASTI, MAHARAJGANJ

### 1. INTRODUCTION

Cupboards are amongst the most important pieces of furniture in a household. It serves to be major means of storage for storing the wearable's including clothing and accessories. The wooden cupboard design holds various assortments of single door, double door, triple door cupboards and may be more. Uttar Pradesh is known for furniture works, Maharajganj and Shravasti are important centres of furniture manufacturing and are home to various units involved in wood working. There are various types of cupboards like built-in cupboard, linen cupboard, stationary cupboard & others.

The term cupboard was originally used to describe an open-shelved side table for displaying dishware, more specifically plates, cups and saucers. These open cupboards typically had between one and three display tiers, and at the time, a drawer or multiple drawers fitted to them. The word cupboard gradually came to mean a closed piece of furniture. It can be used for multiple purposes. Some cupboards are open and doors can be fixed.

The wood crafts of Saharanpur are renowned the world over and have been given a GI tag, which further enhances their presence in the global markets as a unique product.

### 2. RAW MATERIALS

To make a cupboard, the raw materials used are:

- Timber and Wooden Laminates
- Mineral Density Fibre board (MDF)
- Plywood
- Glue and paints
- Polishes
- Screws
- Nails

### 3. MANUFACTURING PROCESS

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.

. The basic steps in manufacturing cupboards are mentioned below:

### 3.1. Selection of wood

Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen. Contemporary cupboards usually have the frame built in solid wood (comprising of the legs and the other components of the frame), the rest is made using plywood/MDF. This is more an economic measure, keeping in mind the costs of the wooden materials.

### 3.2. Cutting of wood

The chosen wood is cut in specific dimensions as per specified designs. There are no fixed standards globally governing the size of cupboards. The wood must stay in a carefully temperature and humidity-controlled room or the wood may swell (too much humidity) or shrink (very dry) and the piece will have cracks when finished. Temperatures must stay in the range of 50-85°F (10-29°C). Contemporary cupboards employ the use of wooden members for purpose of construction of the frame only, with the shutters, sides and back often being substituted with plywood.

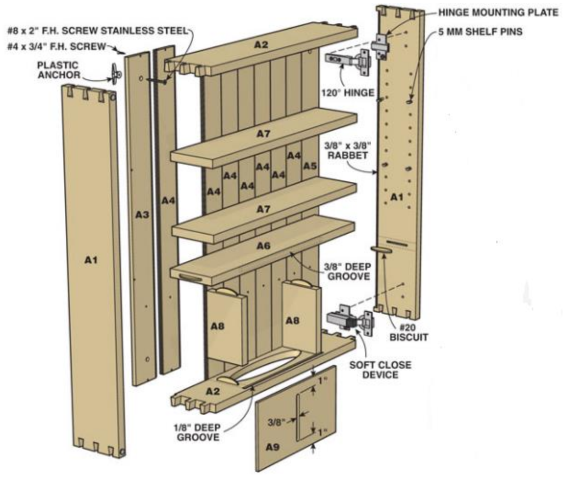


**Figure 1:** Wooden Cupboard

### 3.3. Frame building

The frame is built as per desired size and design and assembled using various joinery details. In the case of the furniture requirement being of a complete wooden cupboard or one made from plywood, wooden slats or plywood are installed as detailed in Figure.1, while for a cupboard constructed with plywood instead of the wooden slats, sections of plywood are installed instead.

### 3.4. Dividers

After frame is put together, dividers are added as observed in the figure above.

### 3.5. Drawers

Drawers are fabricated and attached using channels as per the requirement.

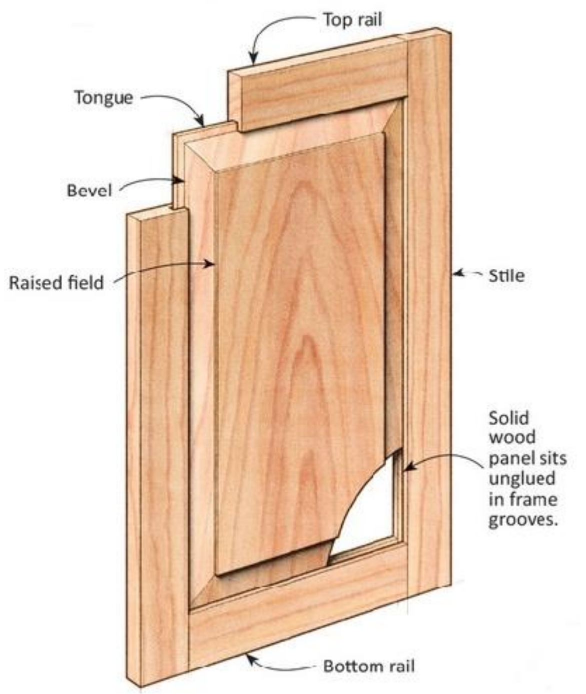


**Figure 2:** Wooden Door construction

### 3.6. Doors

The door frame is fabricated from wooden slats and grooves are cut on the inner faces. The bottom and two side door frame members are attached together using various wooden joinery details. The solid wooden panel having tongues cut onto the edges, is then inserted into the wooden frame. After the wooden panel has been installed, the top rail is installed, completing the wooden door. The doors may comprise of more than one panel, or may have panels of glass.

### 3.7. Finishing

Finally, the cupboard is then checked to remove dust, saw dust and unwanted particles. The cleaned cupboard is then polished. In case of cupboards manufactured using a wooden frame, veneer sheets may be installed on the surface of the plywood and polished to imitate the grains of wood.

### 3.8. Fitting hardware

Hardware such as handles, locks etc. are then installed for final use.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The furniture should be well polished with the polish and varnish applied being uniform, devoid of patches
- The furniture should be free from cracks.
- The joints should be tight.
- The wood should be dry and free from any kind of termites or borers.
- The edges should be rounded.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)

- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>b. European Union Timber Regulation (EUTR)<br>c. General Product Safety Directive (GPSD) | a. European Commission<br>b. European Chemical Agency (ECHA) | a. Legality of timber<br>b. Traceability<br>c. Procedural framework<br>d. Marking and labelling |
| 2 | United States of America | a. The Lacey Act,2008<br>b. California Proposition 65, Act of 1986<br>c. Bureau of Consumer Protection Regulations<br>d. Environment Protection Act (EPA)<br>e. Consumer Product Safety Improvement Act (CPSIA), 2008 | a. Consumer Product Safety Commission (CPSC)<br>b. Animal and Plant Health Inspection Service (APHIS)<br>c. United States Environment Protection Agency (EPA)<br>d. California State Government<br>e. Federal Trade Commission | a. Nature of Timber (endangered or not)<br>b. The genus and species of tree<br>c. Quantity of regulated material<br>d. Country of origin<br>e. Formaldehyde in wood<br>f. Labelling<br>g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act 2010<br>c. Illegal Logging Prohibition Act 2012 | a. Australian Competition and Consumer Commission Act (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations | a. Safety, Saudi Standards, Metrology and Quality | a. General product safety regulation<br>b. Labelling Requirement |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | **b.** SASO Certificate of Conformity | Organization (SASO) | |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Paints and related materials | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood work | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **GSO 450: 1994** | Specifications for paints and varnishes |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes - Specification | USA | **ASTM - D4761** | Test method for wood based material |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | European Union | **ISO/DIS 9665** | Specification for adhesives — Animal glues |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue for general wood working purposes | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 12406: 2021** | Medium density fibre boards for general purpose | USA | **ASTM D143** | Test methods for timber |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture and cabinets- Classification | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | Australia | **AS 2796** | Methods related to seasoning of timber |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber under different wood working operations | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| | **IS 6760: 1972 (Reaffirmed Year: 2021)** | Slotted counter sunk head wood screws | Middle East | **GSO 1424: 2002** | Analysis of moisture content in timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Code of practice | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | European Union | **ISO 3350: 1975** | Timber hardness |
| | **IS 4116: 1988 (Reaffirmed Year: 2017)** | Wooden shelving cabinets (adjustable type) -Specification | European Union | **ISO 21887: 2007** | Timber durability**:** Use classes |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | Middle East | **GSO ISO 21887: 2007** | Timber durability**:** Use classes |
| | - | - | Middle East | **GSO CEN/TS 16818** | Durability of wood and wood-based products - Moisture dynamics of wood and wood-based products |
| | - | - | European Union | **ISO 4211 - 2: 2013** | Test for surface finishes on wooden surfaces |
| | - | - | European Union | **ISO 7170: 2021** | Furniture — Storage units — Determination of strength and durability |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of Terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ Botanical features and other purely scientific terms are excluded from the scope of this standard |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practice for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paint | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Glossary of terms in timber-related Standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Materials | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes -Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of600-900 kg/m3.<br>▪ This standard does not cover veneered or laminated or pre-laminated or other specially treated boards, moulded boards, etc. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture and cabinets-Classification | ▪ This standard covers the general classification of Indian Timber species suitable for furniture and cabinets. It also lays down the general requirements of quality, moisture content and preservative treatment for the timber. |
| | **IS 6760: 1972 (Reaffirmed Year: 2021)** - Slotted countersunk head wood screws | ▪ Requirement and specifications of screws used in wooden furniture |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian standards, namely, IS 7315 : 1974, IS 287: 1993 (Reaffirmed Year: 2017) : 1993 and IS 11215 :1991 respectively. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning,<br>b. Sanding,<br>c. Turning,<br>d. Shaping,<br>e. Mortising, and<br>f. Boring.<br>▪ The common wood working operations used in the manufacture of timber stores and products. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw material | **ASTM - D4761** - Test method for wood- based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/DIS 9665** - Adhesives - Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM 3618: 2005: R2015** - Paint - Detection of lead | ▪ Detection of lead in paints |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lb.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of testing of timber part 1: Standard methods of testing small clear specimens of timber |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 3 Resistance to marking by oils and fats<br>▪ Methods of test for clear finishes for wooden furniture: Part 4 Resistance to marking by liquids |

| Stage | Code | Scope |
|---|---|---|
| | | ▪ Methods of test for clear finishes for wooden furniture: Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI: Resistance to mechanical damage |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 4116: 1988 (Reaffirmed Year: 2017)** -Wooden shelving cabinets (adjustable type) - Specification | ▪ Requirements for materials, sizes, construction and dimensions of wooden cabinets with doors. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Part 1: Operations and workmanship<br>▪ Part 2: Schedules |

### 6.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 3350: 1975** - Wood - Determination of static hardness | ▪ Specifies a method for the determination of the static hardness of wood. |
| | **ISO 21887: 2007** - Durability of wood and wood-based products - Use classes | ▪ Methods to analyse the durability of the wood products |
| | **GSO ISO 21887: 2007** - Durability of wood and wood-based products -Use classes | ▪ Methods to analyse the durability of the wood products |
| | **GSO CEN/TS 16818** - Durability of wood and wood-based products - Moisture dynamics of wood and wood-based products | ▪ Specifies methods to determine the moisture content in wood and wood-based materials. |

| Stage | Code | Scope |
|---|---|---|
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


1. Product information- http://odopup.in/en/article/shravasti
2. Product information- http://odopup.in/en/article/maharajganj
3. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
4. ISO Standards: https://www.iso.org/obp/ui#search
5. EU: https://www.en-standard.eu/
6. USA : https://www.en-standard.eu/astm-standards/, https://www.astm.org/
7. AUS : https://www.standards.org.au/
8. GULF: https://www.gso.org.sa/store/
9. NABL accredited laboratories : https://nabl-india.org/, https://parakh.ncog.gov.in
10. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
11. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

# JAALI PARTITION AND PHOTO & MIRROR FRAME, SAHARANPUR

## 1. INTRODUCTION

Popularly known as the 'Sheesham wood village', Saharanpur is home to some of India's finest wood carvers and is known for its wood work notably, Jaali partition (room divider), mirror and Photo frames.

Jaali partition (room divider) is a screen or piece of furniture placed in a way that divides a room into separate areas. Room dividers are used by interior designers and architects as means to divide space into separate distinct areas. There are a number of different types of room dividers such as cubicle partitions, shoji screens, and walls. Room dividers can be made from various materials, including wood, fabric, framed cotton canvas, or mirrors.

A frame is a protective and decorative edging for a picture or mirror. It makes displaying the work safer and easier and both set the picture/mirror apart from its surroundings or aesthetically integrate it with them. Similarly, mirror frame is a protective and decorative edging for a mirror.

The wood crafts of Saharanpur are renowned the world over and have been given a GI tag, which further enhances their presence in the global markets as a unique product.

## 2. JAALI PARTITION

A room divider is a screen or piece of furniture placed in a way that divides a room into separate areas. Room dividers can be made from many materials, including wood.

In general, room dividers are used in one of these ways:

- To divide rooms, creating a more efficient use of the space within the room.
- As decorators and/or accent pieces to add character to room space.

- To hide areas of different usage or privacy protection
- To decorate rooms for better home design effect

Room dividers also differ in nature being either:

- Permanent as in using wine shelves in restaurants
- Built in as in sliding partitions in offices
- Portable or temporary as for example in convention centers
- Fixed room dividers and hanging room dividers

Popularly known as the 'Sheesham wood village', Saharanpur is home to some of India's finest wood carvers. The city is internationally famous for this craft and its artisans who have been creating magic with the material for years. They 'breathe life into dead trees' is how they like to put it. The intricate and fine workmanship is what makes these products unique. The vine-leaf patterns are a specialty of this region. Geometric and figurative carving is also done along with absolutely gorgeous brass inlay work.

## 3. PHOTO & MIRROR FRAME

Traditionally, frames have been made of wood, and it remains very popular because wood frames can provide strength, be shaped in a broad range of profiles, and allow a variety of surface treatments.

Frames protect the mirror/painting or any other mounted art and give a decorative edge to it. Traditionally, Photo/mirror frames have been made of wood, and it remains very popular because wood frames can provide strength, be shaped in a broad range of profiles, and allow a variety of surface treatments. Frames are generally square or rectangular, though circular and oval frames are not uncommon. Frames in more unusual shapes such as stars, heart can be hand carved by a professional wood carver or carpenter (or possibly melded out of wood pulp). There are also mirror frames designed to go around corners. It is generally made of teak wood.

## 4. RAW MATERIALS

To make a Jali screen or a photo frame, the raw materials used are:

- Timber
- Plywood
- Glue and paints
- Polishes

## 5. MANUFACTURING PROCESS JALI PARTITIONS

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.

A wide range of wood types is used to make products. All of them go through the same basic steps - slicing, carving, inlaying, sanding, polishing and assembling. Craftsmen specialize in one (or more) of these processes and perform just that every day. So, each product is actually handcrafted by not one workman (karigar) but many.

The basic steps in manufacturing jaali partition are:

### 5.1. Slicing of the wood

The first step is to cut the logs of wood into flat sections using an electric saw. Usually, two men hold the wood firmly to cut it precisely. These pieces are marked with pencil and cut in different shapes and sizes according to the desired product, then are worked upon by the carving artisans. For larger Jali partition's it is common for multiple panels to be attached to each other. Contemporary jali partitions may also utilize MDF panels, which can be expertly cut in a wide range of intricate patterns by Computer Numerical Control (CNC) machines. These panels are then mounted between the frames constructed of wood.

### 5.2. Carving

Wood Carving is done either completely by hand or in combination with an electric saw. The saw is usually employed to make geometric patterns in wooden products. The whole operation starts by drawing the pattern intended to be carved on a piece of paper and pasting it over the surface. Holes are drilled to precisely (and quickly) carve out the negative spaces. Though the instruments used are extremely simple (saw, plane, fine-grained hard stone, chisels etc.), the carvers use them aptly to produce some astonishing results with minute details, keeping the intricacies and subtle light and shade effects, every desired curve, expression and texture. These may also be without any perforations, used as dividers.

**Figure.1:** A wooden Jali Partition

### 5.3. Inlaying

Brass is extensively used for inlaying floral, geometric and typographic patterns into wood. This is done by cutting strips of metal and then die-pressing them to get the desired shape. These pieces are then sold by weight to the carvers who make grooves for them beforehand. They beat these into the wood; stick them with wood adhesive and finally nail for durability. The waste strips are recycled into fresh metal and used again.

### 5.4. Polishing

Once the wood is smoothened by rubbing repeatedly using emery paper of different grades, polishing experts come into the picture and putty is applied to even out the unfinished edges and cracks. When it dries, the wooden piece is smoothened further with sandpaper and then coated with paint or polish, depending on the desired finish.

### 5.5. Assembling

Adding rivets, metal hinges and other interesting elements is about giving the product its final touch and assembling several components is just the last task before it is going into its packaging box.

- **General Quality Checks:** Following are a few visual quality parameters/checks:

  a. Polish and varnish applied should be uniform, devoid of patches

  b. The Jali should not have rough unfinished edges

  c. The joints should be tight.

## 6. MANUFACTURNG PROCESS FRAMES

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.

  A wide range of wood types is used to make products. All of them go through the same basic steps - slicing, carving, inlaying, sanding, polishing and assembling. Craftsmen specialize in one (or more) of these processes and perform just that every day. So, each product is actually handcrafted by not one workman (karigar) but many.

The basic steps in manufacturing frames are:

### 6.1. Slicing of the wood

The first step is to cut the logs of wood into flat sections using an electric saw. Usually, two men hold the wood firmly to cut it precisely. These pieces are marked with pencil and cut in different shapes and sizes according to the desired product, then are worked upon by the carving artisans. The frames are always custom made depending on the size of the painting or mirror. The customer conveys the size required to the artisan and the wood sections are cut accordingly.

### 6.2. Framing

The wooden sections are cut as per the sizes required and grooves cut into the inner sides to create recesses for the painting or mirror to rest. The craftsmen use small dowels to ensure a sure joint between the members of the frame.

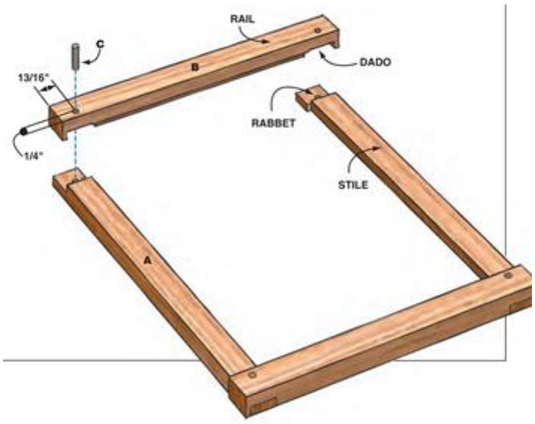


**Figure.2:** Frame assembly

### 6.3. Carving

Based on the design preference the artisans take to decorating each member of the frame, which can be intricately carved in a wide variety of designs, only limited by the ingenuity of the craftsman. The designs may be sketched directly on to the wood and carved using basic hand held tools such as chisels and hammers. After the carvings are completed the frames are assembled. They are smoothened of all rough edges by rubbing with emery paper and carefully polished.

### 6.4. Installing the Artwork/Mirror

The frame is placed upside down and the artwork/ mirror is installed and a thin piece of plywood/MDF board of the size of the inner frame is placed over the back of the artwork/mirror and using special clips, the cover is fastened, to prevent the mounted article from falling out of the back.

### 6.5. Hanging the Frame

Special hardware is installed on the upper member of the frame to allow hanging securely. This enables the mounted artefact to be hung securely.

- **General Quality Checks:** Following are a few visual quality parameters/checks:
  - **a.** The frames should be well polished.
  - **b.** The frames should be free from cracks.
  - **c.** The joints should be tight.
  - **d.** The wood should be dry and free from any kind of termites or borers.
  - **e.** The edges should ideally be rounded.

## 7. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 7.1. Domestic:

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 7.2. International:

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | **a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>**b.** European Union Timber Regulation (EUTR)<br>**c.** General Product Safety Directive (GPSD) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Legality of timber<br>**b.** Traceability<br>**c.** Procedural framework<br>**d.** Marking and labelling<br>**e.** Hazardous chemicals |
| 2 | United States of America | **a.** The Lacey Act,2008<br>**b.** California Proposition 65, Act of 1986<br>**c.** Bureau of Consumer Protection Regulations<br>**d.** Environment Protection Act (EPA)<br>**e.** Consumer Product Safety Improvement Act (CPSIA),2008 | **a.** Consumer Product Safety Commission (CPSC)<br>**b.** Animal and Plant Health Inspection Service (APHIS)<br>**c.** United States Environment Protection Agency (EPA) | **a.** Nature of Timber (endangered or not)<br>**b.** The genus and species of tree<br>**c.** Quantity of regulated material<br>**d.** Country of origin<br>**e.** Formaldehyde in wood<br>**f.** Labelling |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | | d. California State Government<br>e. Federal Trade Commission | g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act 2010<br>c. Illegal Logging Prohibition Act 2012 | a. Australian Competition and Consumer Commission Act<br>b. Department of Agriculture, Water and Environment | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Product Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 8. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Paints and related materials | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood works | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | classification of commercial timbers and their zonal distribution | Middle East | **GSO 450: 1994** | Paints and varnishes |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes - Specification | USA | **ASTM - D4761** | Test method for wood based material |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | European Union | **ISO/DIS 9665** | Specification of adhesives — Animal glues |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** | Converted timber (coniferous) : Part 1 Light furniture | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue for wood working purposes | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | United States of America | **ASTM D143** | Test methods for timber |
| | **IS 12406: 2021** | Medium density fibre boards for general purposes | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber for wood working operations | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | Australia | **AS 2796** | Seasoning of timber |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: code of practice | - | - | - |
| | **IS 3845: 1966 (Reaffirmed Year: 2020)** | Code of practice for joints used in wooden furniture | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | European Union | **ISO 3350: 1975** | Timber hardness |
| | - | - | European Union | **ISO 21887: 2007** | Timber durability: Use classes |
| | - | - | Middle East | **GSO ISO 21887: 2007** | Timber durability: Use classes |
| | - | - | Middle East | **GSO CEN/TS 16818** | Timber durability: Moisture dynamics |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | - | - | European Union | **ISO 4211 -2: 2013** | Test for surface finishes on wooden surfaces |
| | - | - | European Union | **ISO 7170: 2021** | Furniture — Determination of strength and durability |

### 8.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization. |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practise for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paint | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Glossary of terms in timber-related Standards |

## 8.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Materials | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes -Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of600-900 kg/m3.<br>▪ This standard does not cover veneered or laminated or pre-laminated or other specially treated boards, moulded boards, etc. |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** - Converted timber (coniferous): Part 1-Light furniture | ▪ Covers the coniferous species of timber suitable for manufacture of light furniture and similar works, and specifies the requirements of such timber in converted form |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning,<br>b. Sanding,<br>c. Turning,<br>d. Shaping,<br>e. Mortising, and<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315: 1974, IS 287: 1993 (Reaffirmed Year: 2017) and IS 11215 :1991 respectively |

| Stage | Code | Scope |
|---|---|---|
| | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes |

### 8.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/DIS 9665** - Adhesives — Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM 3618: 2005: R2015** - Paint - Detection of lead | ▪ Detection of lead in paints |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lb.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of Timber Part1: Standard Methods of Testing Small Clear Specimens of Timber |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |

### 8.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 3 Resistance to marking by oils and fats<br>▪ Methods of test for clear finishes for wooden furniture: Part 4 Resistance to marking by liquids<br>▪ Methods of test for clear finishes for wooden furniture: Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI: Resistance to Mechanical Damage |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 3845: 1966 (Reaffirmed Year: 2020)** - Code of practice for joints used in wooden furniture | ▪ Covers the joints to be used in locations in various types of wooden furniture. |

### 8.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |

### 8.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 3350: 1975** - Wood - Determination of static hardness | ▪ Specifies a method for the determination of the static hardness of wood. |
| | **ISO 21887: 2007** - Durability of wood and wood-based products - Use classes | ▪ Methods to analyse the durability of the wood products |
| | **GSO ISO 21887: 2007** - Durability of Wood and Wood-Based Products - Use Classes | ▪ Methods to analyse the durability of the wood products |
| | **GSO CEN/TS 16818** - Durability of wood and wood-based products - Moisture Dynamics of Wood and Wood-Based Products | ▪ Specifies methods to determine the moisture content in wood and wood-based materials. |

| Stage | Code | Scope |
|---|---|---|
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


1. Product information- http://odopup.in/en/article/saharanpur
2. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
3. ISO Standards: https://www.iso.org/obp/ui#search
4. EU : https://www.en-standard.eu/
5. USA : https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
6. AUS : https://www.standards.org.au/
7. GULF: https://www.gso.org.sa/store/
8. NABL accredited laboratories : https://nabl-india.org/ , https://parakh.ncog.gov.in
9. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
10. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

# COFFEE TABLE, SIDE TABLE, DINING TABLE, DECORATIVE SIDE TABLE, BASTI, PILIBHIT, SAHARANPUR, MAHARAJGANJ, SHRAVASTI

## 1. INTRODUCTION

A table is considered also a key part of the furniture in any household or commercial set ups. These tables come in various size & shapes serving different purposes. Basti, Saharanpur, Pilibhit, Maharajganj and Shravasti districts of Uttar Pradesh are known for furniture works including different types of decorative wood works.

Tables can be used as side tables, coffee tables, dining table, centre table, etc. based on the requirement. Tables includes top surfaces of various shapes, including rectangular, square, rounded, semi- circular or oval. Legs arranged in two or more similar pairs. However, some tables have three legs and some may use a single heavy pedestal. The table is usually made of hard wood to get a better durability and smooth finishing. Types of woods generally used in India are: Teakwood, sal wood and rosewood.

A few common dimensions of the tables according to their shapes are given as per:

- **Standard Round Table-** 36 to 44-inch diameter
- **Standard Square Table-** 36 to 33-inch square
- **Standard oval Table -** 36-inch diameter, 56 inches' long
- **Standard Rectangular Table-** 36 inches wide, 48 inches' long
- **Standard Dining Table-** 36-40 width, 28-32 height, length as per requirement

The wood crafts of Saharanpur are renowned the world over and have been given a GI tag, which further enhances their presence in the global markets as a unique product.

## 2. RAW MATERIALS

For making tables, the raw materials used are:

- Wooden laminates
- Timber
- Glue
- Paints and polishes
- Screws and nails

## 3. MANUFACTURING PROCESS

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting while in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength.

  The basic steps in manufacturing different types of tables are:

### 3.1. Selection of wood

Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen. Contemporary tables usually have the frame built in solid wood (comprising of the legs and the other components of the frame), the top is made using plywood/MDF. This is more an economic measure, keeping in mind the costs of the wooden materials.

### 3.2. Cutting of wood

The chosen wood is cut in specific dimensions. These sizes are usually standard for tables based on their seating capacity for eg. A six seater table, a four seater table etc. Whereas for side tables the sizes are usually dependent on the customer's choice. The wood must stay in a carefully temperature and humidity-controlled room or the wood may swell (too much humidity) or shrink (very dry) and the piece will have cracks when finished. Temperatures must stay in the range of 50-85°F (10-29°C).

### 3.3. Frame building for table top

The frame is built as per desired size and design and assembled using various joinery details. Apron boards (a panel of wood connecting the table top to the legs of the table) are installed.

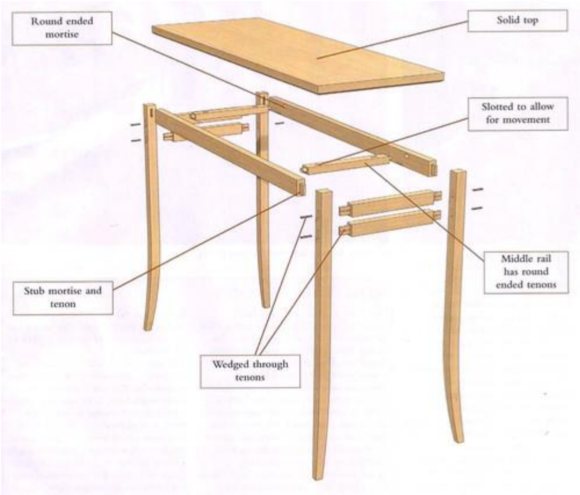


**Figure.1**: Exploded view of table

### 3.4. Rubbing

The wooden frame is rubbed to remove extreme roughness.

### 3.5. Carving

Apart from the process of designing the shape of the furniture, additional decorations in the form of carving on the furniture are done. The Artisans of Uttar Pradesh are adept at carving intricate designs into the wooden furniture. The designs are traced either free hand or by means of a template and using various wooden tools, the craftsmen carve out the wood, leaving behind the intricate design.

### 3.6. Colouring and Protective coating

The table is coloured to match the grains or create textures and a protective coating is applied. At times wooden laminates to mimic the natural grains of wood may be pasted on the table top.

### 3.7. Finishing

Finally, the table is checked to remove dust, saw dust and unwanted particles. The cleaned table is then polished. Veneer may be applied, as required.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The furniture should be well polished.
- The furniture should be free from cracks.
- The joints should be tight.
- The wood should be dry and free from any kind of termites.
- The edges should not be sharp.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)

- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | **a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulations<br>**b.** European Union Timber Regulation (EUTR)<br>**c.** General Product Safety Directive (GPSD) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Legality of timber<br>**b.** Traceability<br>**c.** Procedural framework<br>**d.** Marking and labelling<br>**e.** Hazardous chemicals |
| 2 | United States of America | **a.** The Lacey Act,2008<br>**b.** California Proposition 65, Act of 1986<br>**c.** Bureau of Consumer Protection Regulations<br>**d.** Environment Protection Act (EPA)<br>**e.** Consumer Product Safety Improvement Act (CPSIA), 2008 | **a.** Consumer Product Safety Commission (CPSC)<br>**b.** Animal and Plant Health Inspection Service (APHIS)<br>**c.** United States Environment Protection Agency (EPA)<br>**d.** California State Government<br>**e.** Federal Trade Commission | **a.** Nature of Timber (endangered or not)<br>**b.** The genus and species of tree<br>**c.** Quantity of regulated material<br>**d.** Country of origin<br>**e.** Formaldehyde in wood<br>**f.** Labelling<br>**g.** Children product certificate testing for children furniture |
| 3 | Australia | **a.** Bio Security Import Conditions System (BICON)<br>**b.** Australian Competition and Consumer Commission Act, 2010<br>**c.** Illegal Logging Prohibition Act, 2012 | **a.** Australian Competition and Consumer Commission Act (ACCC)<br>**b.** Department of Agriculture, Water and Environment | **a.** Legality of Logs<br>**b.** Quality of product<br>**c.** Pest free product |
| 4 | Middle East | **a.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>**b.** SASO Certificate of Conformity | **a.** Safety, Saudi Standards, Metrology and Quality Organization (SASO) | **a.** General product safety regulation<br>**b.** Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
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| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Paints and related materials | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Middle East | **GSO 450: 1994** | Paints and varnishes |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood works | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 4116: 1974 (Reaffirmed Year: 2017)** | Glossary :Wooden furniture and fixtures | Australia | **AS 4491:1997** | Terms and specification of terms related to timber |
| Raw Materials (Refer Table 2.a & 2.b) | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes - Specification | European Union | **ISO/DIS 9665** | Specification of animal glue |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue | Middle East | **GSO ISO/TC 218** | Classification of seasoned timber |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | **IS 12406: 2021** | Medium density fibre boards for general use | Australia | **AS 2796** | Seasoning of timber |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture's and cabinets-Classification | Australia | **AS 1080.2.4 - 1981** | Moisture content in timber |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber | USA | **ASTM D143** | Test methods for timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Seasoning of timber: Code of practice | Middle East | **GSO 1424: 2002** | Methods of testing timber |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| Manufacturing/ Processing (Refer Table 3) | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Code of practice | - | - | - |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** | Converted timber (coniferous) : Part 1 Light furniture | - | - | - |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | - | - | - |
| | **IS 3845: 1966 (Reaffirmed Year: 2020)** | Code of practice for joints used in wooden furniture | - | - | - |
| | **IS 4116: 1990 (Reaffirmed Year: 2017)** | Specification of all wooden tops | - | - | - |
| Finishing (Refer Table 4.a & 4.b) | **IS 17633: 2022** | Dining Tables: specifications | European Union | **ISO 7172: 1988** | Methods to determine stability of table |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes - | USA | **ISO 7172: 1988** | Methods to determine stability of table |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | Australia | **AS ISO 7172: 1988: 1988** | Methods to determine stability of table |
| | **IS 5967: Part 1: 1988 (Reaffirmed Year: 2017)** | Method of test for stability determination | Middle East | **GSO ISO 7172: 1988: 2016** | Methods to determine stability of table |
| | **IS 5967:1990 (Reaffirmed Year: 2017)** | School furniture: table, chair | European Union | **ISO 4211 -2: 2013** | Test for surface finishes on wooden surfaces |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | European Union | **ISO 7170: 2021** | Determination of strength and durability of furniture |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ Botanical features and other purely scientific terms are excluded from the scope of this standard. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practise for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paint and varnishes – Terms and definitions | ▪ Details of paints which can be used on wooden surfaces |
| | **IS 4116: 1974 (Reaffirmed Year: 2017)** - Glossary of terms related to wooden furniture and fixture | ▪ Vocabulary and details about terms related to wooden furniture |
| | **AS 2310** - Glossary of paints and painting terms. | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **GSO ISO 1096: 2021** - Specification for plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Terms in timber-related standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes - Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of600-900 kg/m3.<br>▪ This standard does not cover veneered or laminated or pre laminated or other specially treated boards, moulded boards, etc. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber**:**<br>a. Planning<br>b. Sanding<br>c. Turning<br>d. Shaping<br>e. Mortising<br>f. Boring.<br>▪ The common wood working operations used in the manufacture of timber stores and products. |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315: 1974, IS 287: 1993 (Reaffirmed Year: 2017) and IS 11215 :1991 respectively. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture's and cabinets- Classification | ▪ This standard covers the general classification of Indian timber species suitable for furniture and cabinets. It also lays down the general requirements of quality, moisture content and preservative treatment for the timber. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ISO/DIS 9665** - Animal glue | ▪ Specification of glues used in wooden furniture |
| | **ASTM 3618: 2005: R2015** - Paints | ▪ Detection of lead in paints |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |

| Stage | Code | Scope |
|---|---|---|
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lbs.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of Timber Part1: Standard Methods of Testing Small Clear Specimens of Timber |
| | **GSO ISO/TC 218** -Timber | ▪ Standardization of round, sawn and processed timber, and timber materials in and for use in all applications, including terminology, specifications and test methods. |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** -Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** - Converted timber (coniferous) : Part 1 Light furniture | ▪ Covers the coniferous species of timber suitable for manufacture of light furniture and similar works, and specifies the requirements of such timber in converted form (planks and scantlings) |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 3845: 1966 (Reaffirmed Year: 2020)** - Code of practice for joints used in wooden furniture | ▪ Confirmation and practises about the joinery work |
| | **IS 4116: 1990 (Reaffirmed Year: 2017)** - Wooden table tops - Specification | ▪ This standard covers the requirements of materials sizes, construction and finish of board wooden table tops for office, domestic and industrial use |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |
| | **IS 5967: Part 1: 1988 (Reaffirmed Year: 2017)** - Methods of test for strength and stability of tables and trolleys – Stability Strength | ▪ Methods of test for the determination of stability and strength of tables and trolleys. |
| | **IS 5967:1990 (Reaffirmed Year: 2017)** - School furniture, classroom chairs and tables - Recommendations | ▪ Deals with the dimensional requirements of the chairs and tables for children in the age group of 5-17 years for use in the Indian schools. The chairs and tables have been divided into four sizes related to age groups of children.<br>▪ To define a minimum space under the table that shall be kept clear for the knee space<br>▪ To define minimum table lengths based on width between elbows of children when writing, to ensure adequate working space.<br>▪ To provide data on the reach and stretch of (children) arms to assist in the design of working surfaces, any storage space', inkwell positions, etc. |
| | **IS 17633: 2022** - Dining Tables | ▪ Requirements for materials, sizes and functional dimensions of all types of dining tables. |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 3 Resistance to marking by oils and fats<br>▪ Methods of test for clear finishes for wooden furniture: Part 4 Resistance to marking by liquids<br>▪ Methods of test for clear finishes for wooden furniture: Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI: Resistance to mechanical damage |

### 6.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 7172: 1988** -Furniture tables: determination of stability | ▪ Methods for determining the stability of all kinds of tables, except tables permanently attached to the structure of the building. |
| | **ISO 7172: 1988** - Furniture tables: determination of stability | ▪ Methods for determining the stability of all kinds of tables, except tables permanently attached to the structure of the building. |

| Stage | Code | Scope |
|---|---|---|
| | **AS ISO 7172: 1988: 1988** - Furniture tables: determination of stability | ▪ Methods for determining the stability of all kinds of tables, except tables permanently attached to the structure of the building. |
| | **GSO ISO 7172: 1988: 2016** - Furniture tables: determination of stability | ▪ Methods for determining the stability of all kinds of tables, except tables permanently attached to the structure of the building. |
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Part of ISO 4211 specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


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2. Product information- http://odopup.in/en/article/basti
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4. Product information- http://odopup.in/en/article/maharajganj
5. Product information- http://odopup.in/en/article/Raebareli
6. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
7. ISO Standards: https://www.iso.org/obp/ui#search
8. EU : https://www.en-standard.eu/
9. USA : https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
10. AUS : https://www.standards.org.au/
11. GULF: https://www.gso.org.sa/store/
12. NABL accredited laboratories : https://nabl-india.org/ , https://parakh.ncog.gov.in
13. Domestic Legislation/Regulation :
    https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
14. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

# SOFA & DIVAN, BASTI, RAEBARELI, MAHARAJ GANJ, SHRAVASTI, SAHARANPUR

## 1. INTRODUCTION

A couch, also known as a sofa, is a piece of furniture for seating multiple people. It is commonly found in the form of a bench, with upholstered armrests, and often fitted with springs and tailored cushions. Although a couch is used primarily for seating, it may be used for sleeping. In homes, couches are normally put in the family room, living room or lounge. They are sometimes also found in non-residential settings such as hotels, lobbies of commercial offices, waiting rooms, and bars. Couches can also vary in size, colour, and design.

A *sofa* is a piece of furniture for seating two or three people. It is commonly found in the form of a bench, with upholstered armrests, and often fitted with springs and tailored cushions. Although a sofa is used primarily for seating, it may be used for sleeping.

The most common types of sofas include the two-seater, sometimes referred to as a loveseat. The loveseat is designed for seating two people, while some sofa has more than two cushion seats. A sectional sofa, often just referred to as a "sectional", is formed from multiple sections (typically two, three, and four) and usually includes at least two pieces which join at an angle of 90 degrees or slightly greater. Sectional sofas are used to wrap around walls or other furniture. A sofa consists of the frame, padding and covering. The frame is usually made of wood, but can also be made of steel, plastic or laminated boards. Sofa padding is made from foam, down, feathers, fabric or a combination thereof. The covering can be made of fabric, leather, etc.

Another piece of a furniture with a similar purpose, 'divan' is a piece of couch-like sitting furniture or, in some countries, a box-spring based bed. Primarily, in the Middle East, a divan was a long seat formed of a mattress laid against the side of the room, upon the floor or upon a raised structure or frame, with cushions to lean against. The meaning of the word, divan "long, cushioned seat" is due to such seats having been found along the walls in Middle Eastern council chambers.

A *divan* is a piece of couch-like sitting furniture or, in some countries, a box-spring based bed. The divan base is constructed from a sturdy wooden frame surrounded by fabric. It can include drawers and a headboard. The base is then paired with a mattress. The divan base and the mattress are designed to work together to provide a comfortable night's sleep.

Basti, Raebareli, MaharajGanj, Saharanpur and Shravasti are know well known for furniture manufacturing including sofas & divans.

## 2. RAW MATERIALS

The raw materials used for making sofa & divan are:

- Wood (Sagwan, Mango)
- Medium Density Fibre board (MDF)
- Plywood
- Glue and paints
- Polishes
- Dowel
- Nails and screws

## 3. MANUFACTURING PROCESS

- **Seasoning of Wood:** Seasoning is the process of drying timber to remove the moisture contained in walls of the wood cells to produce seasoned timber. Seasoning can be achieved in any number of ways, but the main aim is to remove water at a uniform rate through the piece to prevent damage to the wood during drying (seasoning degrade).

  Seasoned timber tends to have superior dimensional stability than unseasoned timber and is much less prone to warping and splitting in service. In higher grades of timber, particularly hardwoods, the process of seasoning can enhance the basic characteristic properties of timber, increasing stiffness, bending strength and compression strength. Apart from wood seasoning, preservative treatment to wood is also an important procedure to improve the quality, and durability of wood. It enhances the service life of products by 5-10 times. Based on this only many multinational brands claim for life time warranty of their wooden products.

  The basic steps in manufacturing different types of tables are:

### 3.1. Selection of wood

Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen.

### 3.2. Cutting of wood

The chosen wood is cut in specific dimensions. These sizes of sofas are globally recognized into single seater, three seater or a two seater sofa. Divans are usually designed for accommodating three people or again as per the choice of the customer. The wood must stay in a carefully temperature and humidity-controlled room or the wood may swell (too much humidity) or shrink (very dry) and the piece will have cracks when finished. Temperatures must stay in the range of 50-85°F (10-29°C).

### 3.3. Framing

The frame is constructed from wood as per the desired design. The thickness of the wood should allow for the heavy tension webbing to follow. If the frame is not sufficiently strong, it will not bear the weight redistributed into it by the webbing whenever someone sits down. Arms, back, or back sections, seat, and legs are then attached to the main frame. The preferred method is with clean-cut, fitted double doweled glue joints reinforced with comer braces, glued and also screwed into place.

### 3.4. Webbing and springs

The foundation is then set for padding. Jute, a kind of burlap made in India, is used as webbing. Strips of this material are interwoven, stretched across the frame, and tacked down. Flax twine is then used to strap the springs onto the webbing.

### 3.5. Padding

Each part is separately padded as well, with layers of burlap or chosen synthetic material. The padding is placed in a burlap envelope, arranged on the edge of the seat, pinned into place, and stitched down. As the stitching progresses, the pins can be removed one by one. The arms are done next in the same basic fashion. Layers of padding and burlap are fixed in succession and topped with muslin. The arms also have a front edge of extra thick padding. Once the arms are properly shaped the back or back sections may be padded.

### 3.6. Fabric

All the portions of the sofas where the padding is done is covered with desired fabric.

### 3.7. Finishing

Finally, the sofa/divan is then checked to remove dust, saw dust and unwanted particles and given the desired finishing.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The wooden parts should be well polished.
- The wooden parts should be free from cracks.
- The joints should be tight.
- The wood should be dry and free from any kind of termites.
- The edges should be rounded.
- Patterns of fabric and stiches should be uniform.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European

Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>b. European Union Timber Regulation (EUTR)<br>c. General Product Safety Directive (GPSD) | a. European Commission<br>b. European Chemical Agency (ECHA) | a. Legality of timber<br>b. Traceability<br>c. Procedural framework<br>d. Marking and labelling |
| 2 | United States of America | a. The Lacey Act,2008<br>b. California Proposition 65, Act of 1986<br>c. Bureau of Consumer Protection Regulations<br>d. Environment Protection Act (EPA)<br>e. Consumer Product Safety Improvement Act (CPSIA), 2008 | a. Consumer Product Safety Commission (CPSC)<br>b. Animal and Plant Health Inspection Service (APHIS)<br>c. United States Environment Protection Agency (EPA)<br>d. California State Government<br>e. Federal Trade Commission | a. Nature of Timber (endangered or not)<br>b. The genus and species of tree<br>c. Quantity of regulated material<br>d. Country of origin<br>e. Formaldehyde in wood<br>f. Labelling<br>g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act 2010 | a. Australian Competition and Consumer Commission | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | c. Illegal Logging Prohibition Act, 2012 | b. Department of Agriculture, Water and Environment | |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Product Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| General (Refer Table 1) | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Paints and related materials | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood works | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **GSO 450: 1994** | Glossary of terms: Paints and varnishes |
| Raw Materials (Refer Table 2.a & 2.b) | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes - Specification | USA | **ASTM - D4761** | Test method for wood based material |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | European Union | **ISO/DIS 9665** | Specifications of adhesives — Animal glues |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber for wood working purposes | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue for wood working operations | Middle East | **ISO 1096: 2021** | Specification for plywood classification |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | United States of America | **ASTM D143** | Test methods for timber |
| | **IS 12406: 2021** | Medium density fibre boards for general use | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture and cabinets-Classification | Australia | **AS 2796** | Methods for seasoning of timber |
| | - | - | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 5974: 1970 (Reaffirmed Year: 2019)** | Specifications for divans and easy chairs | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Code of practice | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | European Union | **BS EN 13722** | Wooden furniture: Assessment methods |
| | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | European Union | **BS EN 15186: 2012** | Furniture: Assessment on scratching |
| | **IS 12680: 1989 (Reaffirmed Year: 2017)** | Requirements for materials, sizes, construction and dimensions of sofa cum bed used as domestic furniture. | European Union | **BS EN 15187: 2016** | Effect of light on wooden furniture |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood based materials, Part 1: Operations and workmanship | USA | **ASTM D 7023** | Specification: Home furnishing |
| | - | - | USA | **ASTM D 3597** | Classification of woven upholstery fabrics |
| | - | - | Middle East | **GSO ISO 8191 - 1: 2016** | Wooden furniture: Ignitability effect |
| | - | - | European Union | **ISO 4211 - 2: 2013** | Test for surface finishes wooden surfaces |
| | - | - | European Union | **ISO 7170: 2021** | Furniture — Storage units — Determination of strength and durability |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ Botanical features and other purely scientific terms are excluded from the scope of this standard. |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practice for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of Commercial Timbers and Their zonal Distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paint | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Glossary of terms in timber-related standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes - Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of 600-900 kg/m3.<br>▪ This standard does not cover veneered or laminated or pre-laminated or other specially treated boards, moulded boards, etc. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture and cabinets- Classification | ▪ This standard covers the general classification of Indian Timber species suitable for furniture and cabinets. It also lays down the general requirements of quality, moisture content and preservative treatment for the timber. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning<br>b. Sanding<br>c. Turning<br>d. Shaping<br>e. Mortising<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products. |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of Timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315 : 1974, IS 287: 1993 (Reaffirmed Year: 2017) : 1993 and IS 11215 :1991 respectively. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/DIS 9665** - Adhesives — Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |

| Stage | Code | Scope |
|---|---|---|
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lbs.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of timber Part1: Standard methods of testing small clear specimens of timber |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **ASTM 3618: 2005: R2015** - Paint: Detection of lead | ▪ Detection of lead in paints |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for High Density Wood Particle Boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 5974: 1970 (Reaffirmed Year: 2019)** - Specifications for divans and easy chairs | ▪ Refers to domestic furniture and lays down requirements of materials and fixes the functional dimensions of divans |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - permissible Moisture Content For Timber Used For Different Purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 12680: 1989 (Reaffirmed Year: 2017) -** Wooden sofa | ▪ This standard lays down requirements of material, functional dimensions and finishes of wooden sofa-cum-bed used as domestic furniture. |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017) -** Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 2 Resistance to wet heat<br>▪ Methods of test for clear finishes for wooden furniture: Part 3 Resistance to marking by oils and fats<br>▪ Methods of test for clear finishes for wooden furniture: Part 4 Resistance to marking by liquids<br>▪ Methods of test for clear finishes for wooden furniture: Part 5 Test for low angle glare<br>▪ Methods of Test for Clear Finishes for Wooden Furniture - Part VI: Resistance to Mechanical Damage |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020) -** Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |

### 6.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **BS EN 13722** - Furniture: Assessment of the surface gloss | ▪ Specifies a method for the assessment of the surface gloss of furniture surfaces using three reflectometer geometries, 20°, 60° or 85° and relates to rigid surfaces of all finished products regardless of materials, except for finishes on leather and fabrics |
| | **BS EN 15186: 2012** - Furniture: Assessment of the surface resistance to scratching | ▪ Specifies a method for the assessment of the surface resistance to penetrating scratches. It relates to the rigid surfaces of all finished products, regardless of their material. It does not apply to finishes on leather and fabrics. |
| | **BS EN 15187: 2016** - Furniture: Assessment of the surface to light exposure | ▪ Specifies a method for the assessment of the effects of light in indoor conditions, by exposure to artificial radiation and applies to rigid surfaces of all finished products regardless of material. It does not apply to finishes on leather and fabrics. |
| | **ASTM D 7023** - Terminology relating to home furnishings | ▪ Details and specifications of standards related to home furnishing |
| | **ASTM D 3597** - Performance specification for woven upholstery fabrics-Plain, Tufted, or Flocked | ▪ Covers the performance requirements for plain, tufted, or flocked woven upholstery fabrics as used in the manufacture of new indoor furniture. These requirements apply to both the warp and filling directions for those factors where each fabric direction is pertinent. Tests shall be made to determine the properties of the material in accordance with the following test methods: breaking strength; tear strength; resistance to yarn slippage; surface abrasion; dimensional change; colour fastness to water; colour fastness to solvent; colour fastness to burnt gas fumes; colour fastness to crocking; colour fastness to light; colour fastness to ozone; durability of back coating; flammability; |

| Stage | Code | Scope |
|---|---|---|
| | **GSO ISO 8191 - 1: 2016** - Furniture - Assessment of the ignitability of upholstered furniture - Part 1: Ignition source: Smouldering cigarette | ▪ Method of test to assess the ignitability of material combinations, such as covers and fillings used in upholstered seating when subjected to a smouldering cigarette as an ignition source. The tests measure only the ignitability of a combination of materials used in upholstered seating and not the ignitability of a particular finished item of furniture incorporating these materials. They give an indication of, but cannot guarantee, the ignition behaviour of the finished item of furniture |
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


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3. Product information- http://odopup.in/en/article/maharajganj
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5. Product information : https://www.rapportfurniture.com/blogs/rapport-furniture/sofa-dimensions
6. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
7. ISO Standards: https://www.iso.org/obp/ui#search
8. EU : https://www.en-standard.eu/
9. USA : https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
10. AUS : https://www.standards.org.au/
11. GULF: https://www.gso.org.sa/store/
12. NABL accredited laboratories : https://nabl-india.org/
13. Domestic Legislation/Regulation : https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
14. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

# DRESSING TABLE, MAHARAJGANJ, SHRAVASTI

## 1. INTRODUCTION

A dressing table is a piece of furniture often fitted with drawers and a mirror in front of which one sits while dressing and grooming oneself. The natural structure of most of the wooden dressing tables is in the form of a slender wooden plank to which you might have a mirror fixed, a cabinet cum shelf which might have doors or drawers as per the design and a couple of open shelves.

A convenient dressing table is equipped with a mirror of different sizes. A padded leather or a simple seating piece is recommended next to it. In other cases, the item is equipped with a bench for seating. The table top should be wide enough to fit all the necessary cosmetics and various articles of toiletries or other articles of personal grooming.

Structurally, dressing tables are equipped with drawers, one or more for convenient storage of cosmetics, jewellery, perfumes etc. Dressing tables in some styles contain open plan shelves located both above and below the table top.

Different types of Dressing Tables are:

- Sharon dressing table
- Adolph dressing table with storage stool
- Carvel dressing table
- Howler dressing table
- Cinader dressing table

## 2. RAW MATERIALS

Dressing tables are made of a variety of materials such as:

- Timber
- Wooden Laminates and Veneers
- Mineral density fibre board (Medium Density Fibre board (MDF)
- Plywood
- Glue and paints
- Polishes
- Screws and nails

## 3. MANUFACTURING PROCESS

The basic steps in manufacturing different types of tables are:

### 3.1. Selecting the material

The wood is chosen as per the requirement. Different types of wood such as Teak (which has been matured for 50 years or more before felling) are very costly. Also the parameters such as knots, cracks etc. determine the quality of wood and drive its price. Based on the budgetary requirements and quality sought, the wood is chosen. Medium Density Fibre board (MDF) can be used for fabrication at larger volumes. This process is faster as the entire process is mechanized and relatively inexperienced staff can take up production.

### 3.2. Cutting

The material is cut as per the desired dimension and shape. Different parts such as legs, apron boards, drawers etc. are crafted separately and readied for assembly

### 3.3. Rubbing

The wood is rubbed to remove extreme roughness with emery paper of different grades going from coarser to finer till the entire surface of the wood is smooth and even to the feel. For Medium Density Fibre board (MDF) no such rubbing is required. The Medium Density Fibre board (MDF) comes in pre-laminated boards and have a finished texture.

### 3.4. Making Holes and joints

Holes are then drilled for screws and nails. A wide variety of joinery details are used for assembling wooden products such as Tenon and mortise joints, butt joints etc. Once joined they are held in place by wooden dowels which are then sawn off at the top. The whole piece is again sandpapered and the dowel is hammered in and the hole is filled with putty. Medium Density Fibre boards (MDF) are assembled using wooden dowels, small hinges and different types of hardware.

### 3.5. Colouring and Protective coating

The dressing table is coloured and coating is applied to provide resistance to termites etc. No such treatment is required for the modular furniture made using Medium Density Fibre board (MDF).

### 3.6. Legs

Legs are then attached to all sides of the dressing table and final assembly is done.

### 3.7. Finishing

Finally, the table is then checked to remove dust, saw dust and unwanted particles. The cleaned table is then polish is applied as per the demand. The Mirror is attached to the dressing table as a part of the finishing process after the polishing and painting is completed.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The furniture should be well polished.
- The furniture should be free from cracks.
- The joints should be tight and drawer movement should be smooth.
- The wood should be free from termites and borers.
- Mirror should be checked for distorted image.

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>b. European Union Timber Regulation (EUTR)<br>c. General Product Safety Directive (GPSD) | a. European Commission<br>b. European Chemical Agency (ECHA) | a. Legality of timber<br>b. Traceability<br>c. Procedural framework<br>d. Marking and labelling<br>e. Hazardous Chemicals |
| 2 | United States of America | a. The Lacey Act,2008<br>b. California Proposition 65, Act of 1986<br>c. Bureau of Consumer Protection Regulations<br>d. Environment Protection Act (EPA)<br>e. Consumer Product Safety Improvement Act (CPSIA), 2008 | a. Consumer Product Safety Commission (CPSC)<br>b. Animal and Plant Health Inspection Service (APHIS)<br>c. United States Environment | a. Nature of Timber (endangered or not)<br>b. The genus and species of tree<br>c. Quantity of regulated material<br>d. Country of origin<br>e. Formaldehyde in wood<br>f. Labelling |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | | Protection Agency (EPA)<br>d. California State Government<br>e. Federal Trade Commission | g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act 2010<br>c. Illegal Logging Prohibition Act 2012 | a. Australian Competition and Consumer Commission (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General ( Refer Table 1)** | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Glossary terms - Paints and related materials | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Practise for fire safety for industrial use | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | - | - | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Partial boards of wood and other lignocellulose materials (medium density) for general purposes | USA | **ASTM - D4761** | Test method for wood based material |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | European Union | **ISO/DIS 9665** | Specification of adhesives — Animal glues |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber for wood working purposes | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Seasoning of timber: Code of practice | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue | USA | **ASTM D143** | Test methods for timber |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards -Specification | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | USA | **ASTM 4442- 20** | Analysis of moisture content in timber |
| | - | - | Australia | **AS 2796** | Methods for seasoning of timber |
| | - | - | Australia | **AS 1080.2.4 - 1981** | Analysis of moisture content in timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber: Specification | - | - | - |
| | **IS 5247: Part 1: 1982 (Reaffirmed Year: 2020)** | Converted timber (coniferous) : Part 1 Light furniture | - | - | - |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane | - | - | - |
| | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | European Union | **ISO 4211 -2: 2013** | Furniture - Test for surface finishes on wooden surfaces |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | European Union | **ISO 7170: 2021** | Furniture- storage units-determination of strength and durability |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| **General** | **ISO 4618: 2014** - Paints and varnishes — Terms and definitions | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paint and painting terms |

| Stage | Code | Scope |
|---|---|---|
| | **ISO 24294: 2021**- Vocabulary for timber | ▪ Contains the terms and definitions of concepts to establish a multilingual vocabulary of terminology to be applied in forest and wood working spheres.<br>▪ The scope of identification of a tree and of its parts in round and sawn aspects; its measurements; grading; condition; features; sizes; and the natural, biological and infestation defects of wood. |
| | **AS 4491:1997** - Glossary of terms related to timber | ▪ Glossary of terms in timber-related standards |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practice for fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of commercial timbers and their zonal distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes - Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocellulose materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ This Standard does not cover veneered particleboards, moulded particle boards, high and low-density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards - Specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of animal glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** - Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber:<br>a. Planning<br>b. Sanding<br>c. Turning<br>d. Shaping<br>e. Mortising<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315: 1974, IS 287: 1993 (Reaffirmed Year: 2017) and IS 11215 :1991 respectively. |
| | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high- density wood particle boards in flat sheet or moulded forms. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/DIS 9665** - Adhesives — Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lbs.) and the basic SI unit of mass (Kg) are cited. |
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of Timber Part1: Standard Methods of Testing Small Clear Specimens of Timber |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ Standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |

| Stage | Code | Scope |
|---|---|---|
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **ASTM 3618: 2005: R2015** - Paint: Detection of lead | ▪ Detection of lead in paints |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms | ▪ Covers the coniferous species of timber suitable for manufacture of light furniture and similar works, and specifies the requirements of such timber in converted form (planks and scantlings). |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | Methods of test for clear finishes for wooden furniture:<br>▪ Part 1 Resistance to dry heat<br>▪ Part 2 Resistance to wet heat<br>▪ Part 3 Resistance to marking by oils and fats<br>▪ Part 4 Resistance to marking by liquids<br>▪ Part 5 Test for low angle glare<br>▪ Part 6 Resistance to Mechanical Damage |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | ▪ Code of practice for finishing of wood and wood-based materials.<br>▪ Part 1 - Operations and workmanship<br>▪ Part 2 - Schedules |

### 6.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |

| Stage | Code | Scope |
|---|---|---|
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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    c. AUS: https://www.agriculture.gov.au/import/goods/timber

## WOODCRAFT, BIJNOR, VARANASI & CHITRAKOOT

### 1. INTRODUCTION

Wood carving is the technique of creating elaborate designs in wood by hand, with the help of carving tools. Wood carving may vary from traditional motifs to geometrical or abstract patterns. Carving wooden handicrafts is a laborious process as great attention needs to be paid to the minutest of details. Fine woodworking still remains a craft pursued by many. There is an increased demand for hand crafted work such as furniture and arts and artisans are reaching out to embrace technology to gain access to contemporary designs and infusing them in their crafts.

Wood carving in India is a traditional art which is passed down through the generations. Wood carving involves shaping wood to make objects of utility and chiselling parts of wood to form intricate designs, with the help of hand tools. Articles of daily use like rolling pins, ladles, walking sticks, and combs are made from softwoods, while exotic wood like sandalwood, ebony, walnut, rosewood and teak are used to carve items of decorative value.

Craftsmen in Bijnor, Varanasi and Chitrakoot are renowned for their skills in carving a wide range of wooden handicrafts like wood furniture, decorative panels, wooden screens, toys, spoons, bowls, trays, vases, book stands, jewel boxes, window frames, masks, idols, photo frames, key hangers, beads etc.

### 2. PRODUCT

#### 2.1. Wooden Toys

Wooden toy-making is a traditional craft in the Varanasi & Chitrakoot districts of Uttar Pradesh, India. Bright and colourful lacquered toys are made by clusters of skilled craftsmen. These toys from Varanasi were given the Geographical Indication tag in 2014, along with other lacquerware produced in this region.

Wooden logs are sourced from nearby areas such as the jungles of Chitrakoot and Sonbhadra. Keria (Coraiya) wood from Bihar was used earlier and it is still the preferred type of wood for toy-making. But after the government banned the cutting of Keria trees in the 1980s, the craftsmen switched to using the wood from eucalyptus, which is now used predominantly. Stacks of wooden logs can be seen stored outside the houses of the wood-carvers.

a. **Raw Materials:** The primary raw materials used in the manufacture of toys are:

- Eucalyptus wood
- Natural Dyes
- Lac

b. **Manufacturing Process:** The basic steps in manufacturing wooden toys are:

- **Shaping the wood:** The process commences with the selection of a suitable wood for the toy being prepared. The design can be a simple one or a complex one depending on the skill of the artisan. The wood piece is heated evenly over a flame by the master artisan to remove all traces of moisture. Using emery paper, the wood piece is sanded down to obtain a smooth surface. These pieces are mounted on lathe machine and with the help of different types of chisel the wood is shaped in spherical, circular or oval shapes as per the design the form is then rubbed with a sand paper in order to smoothen the surface.
- **Colour preparation:** For the purpose of preparation of colours, the artisans use natural colours, obtained from various vegetable or other natural pigments such as Turmeric, Flowers such as Amaltas, Hibiscus etc. They are dried and ground to obtain the dried colours. These colours are added to hot lacquer and solidified as sticks.
- **Colouring:** The toy is mounted on the lathe and as it turns the solidified lacquer stick is gently pressed on to it. Due to the friction, the lac heats up and melts and gets applied on the toy turning on the lathe.
- **Assembly:** Once the lac application is finished, the product is removed from the lathe and further surface embellishment or assembling of different parts (if the toy is made of 2 or more parts) is done. Other decorative works on the surface of toy are also done at this stage.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- There should be no sharp edges or parts.
- The toy should be free from cracks or damages.
- The colours should be uniformly applied.

### 2.2. Wood carving and sculptures

Wood carving in India has been around since ancient times, with ancient wood carved temples in Uttar Pradesh. The craftsmen from ‘Nagina’ in the district of Bijnor are recognized for their artistic prowess globally and the city itself is known globally as the “wood craft city”. The artisans procure the woods and rely on traditional methods to turn blocks of wood into exquisite works of art. While

Sheesham is the most widely used type of wood, mango, walnut, deodar, ebony and sandalwood are also used. Artistic & architectural woodwork began as a palace & temple craft and over time evolved to daily use artefacts.

a. **Raw Materials:** Seasoned wood logs, polish and paints

b. **Manufacturing Process:** The basic steps in manufacturing wooden carvings and sculptures are:

- **Selection of the wood:** Wood used for carving may be classified into two kinds, namely, hardwood and softwood.

  Hardwood refers to wood that is obtained from deciduous broadleaved trees, which shed their leaves annually. Examples of hardwood trees are teak, oak, rosewood, sandalwood, walnut etc.

  Softwood comes from trees bearing cones, most of which are evergreen. It is ideal for wood carving, since it is easy to cut, doesn't splinter, glues well and is good for laminating. Softwood includes pine, cedar, and fir trees.

- **Carving techniques:** Various techniques to carve wood exist. Wood carving may be broadly classified into four types:

i. **Deep wood carving**: It is usually two inches deep, or more. These carvings are replete with intricate floral and animal motifs. This requires a lot of labour and skill, and is the most expensive form of carving.

ii. **Shallow wood carving**: It is usually half an inch deep. In this form of carving, skilled artisans carve patterns on a flat surface. This form of carving is characterized by mythological themes.

iii. **Latticework**: in wood carving, involves ornate designs which are carved onto the wood. Mostly used for windows, this form of carving portrays carved motifs of interlaced foliage, animals, and birds, besides others.

iv. **Semi-carving**: This is done on a thin panel along the rim of a surface. This form of carving is a carver's delight since it allows the grains of wood to be displayed along with the carver's skills.

These wood carving techniques are then categorised into various wood carving styles such as Whittling, Chip Carving, Relief Carving, Intaglio Carving and Carving in the Round. Now with advent of Computer Numerical Control (CNC) machines, many fine designs are possible.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- The wood carving/sculpture should be well shaped with smooth edges.
- The colours or polish should be uniformly applied without patches.
- The wood should be free from termites and borers.

## 3. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 3.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | REMARKS |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 3.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | **a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulations<br>**b.** European Union Timber Regulation (EUTR)<br>**c.** General Product Safety Directive (GPSD) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Legality of timber<br>**b.** Traceability<br>**c.** Procedural framework<br>**d.** Marking and labelling |
| 2 | United States of America | **a.** The Lacey Act,2008<br>**b.** California Proposition 65<br>**c.** Bureau of Consumer Protection Regulations<br>**d.** Environment Protection Act (EPA) | **a.** Consumer Product Safety Commission (CPSC)<br>**b.** Animal and Plant Health Inspection Service (APHIS) | **a.** Nature of Timber (endangered or not)<br>**b.** The genus and species of tree<br>**c.** Quantity of regulated material<br>**d.** Country of origin |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | e. Consumer Product Safety Improvement Act (CPSIA),2008 | c. United States Environment Protection Agency (EPA)<br>d. California State Government<br>e. Federal Trade Commission | e. Formaldehyde in wood<br>f. Labelling<br>g. Children product certificate testing for children furniture |
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act 2010<br>c. Illegal Logging Prohibition Act 2012 | a. Australian Competition and Consumer Commission (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 4. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 4908: 1968 (Reaffirmed Year: 2018)** | Glossary of terms used in lac industry | - | - | - |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | - | - | - |
| **Raw Materials ( Refer Table 2.a & 2.b )** | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of all types of glue used for woodworking purposes. | - | - | - |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | USA | **ASTM - D4761** | Test method for wood-based material |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Methods to test various types of timber | USA | **ASTM D143** | Test methods for timber |
| | **IS 16: Part 1: 2008 (Reaffirmed Year: 2018)** | Shellac - Part 1: Handmade shellac | - | - | - |
| | **IS 16: Part 2: 2008 (Reaffirmed Year: 2018)** | Shellac: Part 2 Machine-made shellac | - | - | - |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood- based materials, Part 1: Operations and workmanship | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | European Union | **ISO 4211 -2: 2013** | Test for surface finishes. |
| | **IS 9873: Part 1: 2019** | Safety of toys, Part 1: Safety aspects related to mechanical and physical properties (fourth revision) | United States of America | **ASTM D143** | Test methods for timber |
| | **IS 9873: Part 2: 2017** | Safety of toys, Part 2: Flammability (third revision) | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| | **IS 9873: Part 3: 2020** | Safety of toys, Part 3: Migration of certain elements (third revision) | - | - | - |
| | **IS 9873: Part 6: 2021** | Safety of toys Part 6 Determination of certain phthalate esters in toys and children products | - | - | - |
| | **IS 9873: Part 9: 2020** | Safety of toys Part 9 Certain phthalates esters in toys and children's products | - | - | - |

## 4.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| **General** | **IS 4908: 1968 (Reaffirmed Year: 2018)** - Glossary of Terms used in Lac Industry | ▪ Glossary of terms |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms | ▪ Glossary of terms |

## 4.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| **Raw Materials** | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of working quality of timber | ▪ Methods to test various types of timber<br>▪ General safety provisions |

| Stage | Code | Scope |
|---|---|---|
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ Specification of all types of glue used for woodworking purposes. |
| | **IS 16: Part 1: 2008 (Reaffirmed Year: 2018)** - Handmade shellac | ▪ Description of form and mandatory requirements<br>▪ Tests and methods<br>▪ Sampling methods<br>▪ Adjunct Standards and referred documents |
| | **IS 16: Part 2: 2008 (Reaffirmed Year: 2018)** - Machine-made shellac | ▪ Description of form and mandatory requirements<br>▪ Grading, tests and methodology<br>▪ Sampling methods<br>▪ Adjunct standards and referred documents |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of Timber | ▪ Classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo. |

### 4.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood- based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ASTM D143** - Test methods for timber | ▪ Determination of strength of wood<br>▪ Physical and mechanical properties of wood |

### 4.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood- based materials | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |

### 4.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ Recommendations for permissible limits of moisture content for different stores in each of the four climatic zones in the country for the purpose of ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 9873: Part 1: 2019** - Safety of toys, Part 1 Safety aspects related to mechanical and physical properties (fourth revision) | ▪ Safety aspects related to mechanical and physical properties |
| | **IS 9873: Part 2: 2017** - Safety of toys Part 2 flammability (third revision) | ▪ General requirements relating to all toys and specific requirements and methods of test relating to the following toys which are considered as being those presenting the greatest hazard.<br>▪ Prohibited materials |
| | **IS 9873: Part 3: 2020** - Safety of toys Part 3 Migration of certain elements (third revision) | ▪ Maximum acceptable levels and methods of sampling, extraction and determination for the migration of the elements- antimony, arsenic, barium, cadmium, chromium, lead, mercury and selenium from toy materials and from parts of toys |
| | **IS 9873: Part 6: 2021** - Safety of toys Part 6 Determination of certain phthalate esters in toys and children's products | ▪ Method for the determination of di-iso-butyl phthalate (DIBP), di-n-butyl phthalate (DBP), benzylbutyl phthalate (BBP), bis-(2-ethylhexyl) phthalate (DEHP), di-n-octyl phthalate (DNOP), di-iso-nonylphthalate (DINP) and di-iso-decyl phthalate (DIDP) (as specified in Annex A) in toys and children's products |
| | **IS 9873: Part 9: 2020** - Safety of toys Part 9 Certain phthalates esters in toys and children's products | ▪ Maximum acceptable levels of certain phthalate esters in toys and children products, except material not accessible. |

### 4.6. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials. |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading. |
| | **GSO 1424: 2002** -Methods of testing of timber, Part1 - Standard methods of testing small clear specimens of timber | ▪ Methods of Testing of Timber Part1: Standard Methods of Testing Small Clear Specimens Of Timber |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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# KITCHEN ITEMS (WOODEN) & HOUSEHOLD GOODS, PILIBHIT, RAEBARELI

## 1. INTRODUCTION

Wooden kitchen items have been in use since antiquity. They were amongst the earliest used implements used in cooking, serving and dining due to easy availability and affordability. Even in today's contemporary kitchens wooden bowls, scoops, salad forks and many other items made of wood are still being used.

Wooden implements are also used for decorative purposes and are often exaggerated in sizes and shapes

The earliest tools used for cooking were fashioned out of wood. They were slowly replaced by other resistant metal wares which were more suited for cooking due to their ability to withstand heat and ease of cleaning and overall sturdiness. In contemporary times, wooden kitchen wares are used infrequently and are treated more as decorative items.

They are still used for stirring many different kinds of food and beverages, specially, soups and casseroles during preparation, although they tend to absorb strong smells such as onion and garlic. Unlike metal spoons, they can also be safely used without scratching non-stick pans.

## 2. RAW MATERIALS

Raw materials for making kitchen items are cured and seasoned sheesham and mango wood. These are shaped into different forms using a variety of wood carving tools such as gouges, chisels, hammers etc.

## 3. MANUFACTURING PROCESS

The basic steps in manufacturing kitchen ware are:

### 3.1. Sketching the design

The design of the equipment being made is marked on the piece of wood. The designs can be simple and utilitarian in nature or ornate ones depending on the craftsman.

### 3.2. Chiselling

The wood is chiselled out to give shape to the product. The details and patterns are further marked and carved out.

### 3.3. Polishing

The product is rubbed thoroughly with emery paper to make it smooth to the touch. Rough edges are carefully carved out. The entire process of rubbing with emery paper is repeated to get the final finish.

### 3.4. Finishing

The finished utensil is polished using boiled linseed oil.

**Figure. 1:** Wooden piece being cut and shaped for spoon


**Figure. 2:** Finished and decorated Roti Box

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for rough edges and uniform smooth finish
- Check the product for any cracks
- Polishing should be uniform

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation Reference | Regulatory Body | Regulation |
|---|---|---|---|---|
| 1 | India | a. Guidelines on Equipment, Utensils and Containers, FSSAI regulations, 2011. | a. FSSAI | a. Governs use of Food grade utensils/crockery<br>b. All equipment and utensils which come in contact with food. |

## 5.2. International

| S. No. | Country | Regulation Reference | Regulatory Body | Regulation |
|---|---|---|---|---|
| 1 | European Union | **a.** Regulation (EC) No. 1935/2004<br>**b.** Declaration of Compliance (DOC)<br>**c.** REACH (Registration, Evaluation, Authorisation and Restriction of Chemicals) Regulations<br>**d.** Restricted Substances<br>**e.** General Product Safety Directive (GPSD)<br>**f.** Regulation (EU) No. 995/2010<br>**g.** European Union Timber Regulations (EUTR ) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Food Safety requirements for Food contact materials (FCM)<br>**b.** Restricted substances<br>**c.** CE marking<br>**d.** Due Diligence against illegal timber |
| 2 | United States of America | **a.** Code of Federal Regulations Title 21 (21 CFR)<br>**b.** Lacey Act, 2008<br>**c.** California Proposition 65, Act of 1986<br>**d.** Bureau of Consumer Protection Regulations<br>**e.** Consumer Product Safety Improvement Act (CPSIA), 2008 | **a.** USFDA<br>**b.** The United States Department of Agriculture Animal and Plant Health Inspection Service (USDA APHIS)<br>**c.** Consumer Product Safety Commission (CPSC)<br>**d.** Federal Trade commission<br>**e.** California State Government | **a.** Food Safety (FCM)<br>**b.** Wooden Product Regulations<br>**c.** Hazardous substances<br>**d.** Country of origin<br>**e.** Flammability, lead |
| 3 | Australia | **a.** Bio Security Import Conditions System (BICON)<br>**b.** Australian Competition and Consumer Commission Act 2010<br>**c.** Illegal Logging Prohibition Act 2012 | **a.** Australian Competition and Consumer Commission (ACCC)<br>**b.** Department of Agriculture, Water and Environment | **a.** Bio Hazard<br>**b.** Illegal Logging |
| 4 | Middle East | **a.** Regulation of Food Contact Materials in the GCC | **a.** GCC Standardization Organization | **a.** Food Contact material regulations and technical standards |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Code of practice for fire safety of industrial buildings - Saw mills and wood works | European Union | **BS EN 50632-1 - 2015+A1 - 2019** | Electric motor-operated tools. Dust measurement procedure general requirements |
| | - | - | USA | **ASTM D1165** | Standard Nomenclature of Commercial Hardwoods and Softwoods |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 1930: 2003 (Reaffirmed Year: 2019)** | Woodworking tools - Chisels and gouges (Third Revision) | USA | **ISO 2729: 1995** | Woodworking tools — Chisels and gouges |
| | **IS 4873 (Part 1,2): 2008** | Methods of Laboratory Testing of Wood Preservatives against Fungi and Borers (Powder Post Beetles) Part 2 Determination of Threshold Values of Wood Preservatives against Borers (Powder Post Beetles) in Wood and Bamboo (Third Revision) | USA | **ISO 3295:1975** | Narrow Band saw blades for woodworking - Dimensions |
| | - | - | Middle East | **GSO ISO 3295: 2013** | Narrow Band saw Blades for Woodworking - Dimensions |
| **Manufacturing/ Processing (Refer Table 3.a & 3.b)** | **IS 3691: 1990 (Reaffirmed Year: 2019)** | Woodworking machines - Table band sawing machines- Nomenclature and acceptance conditions (Second Revision) | European Union | **BS ISO 13061-5: 2020** | Physical and mechanical properties of wood. Test methods for small clear wood specimens Determination of strength in compression perpendicular to grain |
| | - | - | USA | **ASTM D143** | Standard Test Methods for Small Clear Specimens of Timber |
| | - | - | Middle East | **GSO ISO 3129: 2015** | Wood - Sampling Methods And General Requirements For Physical And Mechanical Testing Of Small Clear Wood |
| | - | - | Middle East | **GSO ISO 3350: 2015** | Wood - Determination Of Static Hardness |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 4833: 1993 (Reaffirmed Year: 2017)** | Methods for field testing of preservatives in wood | USA | **ASTM D6874 - 21** | Standard Test Methods for Non-destructive Evaluation of the Stiffness of Wood and Wood-Based Materials Using Transverse Vibration or Stress Wave Propagation |
| | - | - | USA | **ASTM D3345** | Standard Test Method for Laboratory Evaluation of Solid Wood for Resistance to Termites |
| | - | - | AUSTRALIA | **ISO 13061-5: 2020** | Physical and mechanical properties of wood - Test methods for small clear wood specimens |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **BS EN 50632-1 - 2015+A1 – 2019** - Electric motor-operated tools. Dust measurement Procedure General Requirements | ▪ This European Standard specifies general requirements for the dust measurement of electric motor-operated tools supplied from mains or from batteries. This European Standard applies to those tools with and without dust extraction unit where dust such as mineral dust containing silica or wood dust is expected.<br>▪ Dust is a disperse distribution of solid substances in gases, particularly air, resulting from mechanical processes. According to EN 481, two size categories are to be differentiated: the inhalable dust and the reparable dust fraction. Inhalable dust refers to the entire inhalable fraction of the dust through mouth and/or nose. Reparable dust relates to the fraction of the inhalable dust that can reach the pulmonary alveoli due to its small particle size. |
| | **ASTM D1165 -** Standard nomenclature of commercial hardwoods and softwoods | ▪ Commercial species group names are listed with common tree and botanical names. Commercial names are representative of commercial practice in the United States and Canada. Some foreign species that are used in the United States and Canada are listed in Appendix X1 with their commercial and botanical names.<br>▪ The official common names conform to USFS AH 41. In addition to the official common name for a species, the checklist also lists other names by which the species and the lumber produced from it are sometimes designated. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 6329: 2000 (Reaffirmed Year: 2021) -** Code of practice for fire safety of industrial buildings - Saw mills and wood works | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **IS 1930: 2003 (Reaffirmed Year: 2019)** - Woodworking tools - Chisels and gouges (third revision) | ▪ This specifies the characteristics of chisels and gouges for woodworking |
| | **IS 4873 (Part 1,2): 2008** - Methods of Laboratory Testing of Wood Preservatives against Fungi and Borers | ▪ This standard (Part 1) lays down the method for the laboratory determination of threshold value of wood preservatives against fungi by means of soil block and Koll flask methods.<br>▪ This standard (Part 2) lays down the method for the laboratory testing of wood preservatives against borers (powder post beetles) by exposing the wood/ bamboo samples to both adult and larvae of test insects or keeping them along with material infested with powder post beetles in the laboratory. The test can also be employed to determine the natural durability of different species of wood/bamboo. |
| | **IS 4833: 1993 (Reaffirmed Year: 2017)** - Method of field testing preservatives in wood | ▪ Testing methodology |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ISO 2729: 1995** - Woodworking tools — Chisels and gouges | ▪ This International Standard specifies the characteristics of chisels and gouges for woodworking. |
| | **GSO ISO 3295: 2013** - Narrow band saw blades for woodworking - Dimensions | ▪ This International Standard specifies the widths and the thicknesses of narrow band saw blades for woodworking. It also specifies the pitch of some types of teeth. |

### 6.4. Table 3.a: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3691: 1990 (Reaffirmed Year: 2019)** - Woodworking machines - Table Bands awing machines- Nomenclature and acceptance conditions (Second Revision) | ▪ This International Standard specifies the terminology appropriate to each part of the machine and, with reference to ISO/ R 230, the geometrical test for table band sawing machines and gives the corresponding permissible deviations which apply to machines for general purpose use and normal accuracy.<br>▪ This International Standard deals only with the verification of accuracy of the machine. It does not apply to the testing of the running of the machine (vibrations, abnormal noises, stick-slip motion of the components etc.), nor to its characteristics (speeds, feeds etc.) which should generally be checked before testing accuracy |

### 6.5. Table 3.b: (International Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **BS ISO 13061-5: 2020** - Physical and mechanical properties of wood. | ▪ This document specifies a method for determining the strength in compression perpendicular to grain of wood. |
| | **ASTM D143 -** Standard test methods for small clear specimens of timber | ▪ These test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ These test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading. |
| | **GSO ISO 3129: 2015** - Sampling methods and general requirements for physical and mechanical testing of small clear wood | ▪ This document specifies methods for the extensive and limited sampling of wood, conditioning and preparation of test pieces. It also specifies the general requirements for physical and mechanical testing of small clear wood specimens. The sampling guidance provided in this document can be applied for timber taken from either trees, logs, or pieces of ungraded/graded/pre-sorted sawn timber for non-structural applications, such as furniture, windows, doors, etc., only. |
| | **GSO ISO 3350: 2015** - Determination of static hardness | ▪ Specifies determination of static hardness in wood |

### 6.6. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 4833: 1993 (Reaffirmed Year: 2017)** - Methods for field testing of preservatives in wood | ▪ This standard prescribes the requirements and the methods of sampling and test for KHOA. |

### 6.7. Table 4.b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ASTM D6874 – 21:** Standard test methods for non-destructive evaluation of the stiffness of wood and wood-Based materials using transverse vibration or stress wave propagation | ▪ Hose test methods cover the non-destructive determination of the following dynamic properties of wood and wood-based materials from measuring the fundamental frequency of vibration:<br>▪ Flexural (see Refs (1-3))2 stiffness and apparent modulus of elasticity (Etv) properties using simply or freely supported beam transverse vibration in the vertical direction, and<br>▪ The test methods can be used for a broad range of wood-based materials and products ranging from logs, timbers, lumber, and engineered wood products. |

| Stage | Code | Scope |
|---|---|---|
| | **ASTM D3345:** Standard test method for laboratory evaluation of solid wood for resistance to termites | ▪ This test method covers the laboratory evaluation of treated or untreated solid wood for its resistance to subterranean termites. This test is considered as a screening test for treated material and further evaluation by field methods is required.<br>▪ The values stated in SI units are to be regarded as the standard. The values given in parentheses are for information only.<br>▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO 13061-5: 2020:** Physical and mechanical properties of wood - Test methods for small clear wood specimens | ▪ This document specifies a method for determining the strength in compression perpendicular to grain of wood. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


1. Product information : http://odopup.in/en/article/raebareli
2. Product information : http://odopup.in/en/article/pilibhit
3. Product information : https://en.wikipedia.org/wiki/Wooden_spoon
4. Product information http://brandbharat.com/english/up/districts/Raebareli/Raebareli.html
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6. Product Information : https://www.instructables.com/Make-a-bunch-of-kitchen-utensils-from-a-block-of-w/
7. Domestic Legislation/Regulation : https://fssai.gov.in/cms/food-safety-and-standards-regulations.php
   https://www.forests.tn.gov.in/app/webroot/img/document/legislations/14-Guidelines-WoodBasedIndustries_22092016.pdf
8. International Legislation/Regulation:
   a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
   b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
   c. AUS: https://www.agriculture.gov.au/import/goods/timber
9. Domestic Standards for Food Processing: https://standardsbis.bsbedge.com/
10. ISO Standards: https://www.iso.org/obp/ui#search
11. NABL accredited laboratories: https://nabl-india.org/ , https://parakh.ncog.gov.in
12. International Standards:
    a. EU : https://www.en-standard.eu/ , https://www.bsbedge.com
    b. USA : https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
    c. AUS : https://www.standards.org.au/
    d. GULF: https://www.gso.org.sa/store/

# MODULAR KITCHEN, RAEBARELI

## 1. INTRODUCTION

Kitchen cabinets are the built-in furniture installed in many kitchens for storage of food, cooking equipment and dishes for table service.

Appliances such as refrigerators, dishwashers, and ovens are often integrated into kitchen cabinetry. There are many options for the cabinets available at present.

Modern kitchen design has improved partly as a result of ergonomic research. Functionality is important; one research study had "anthropological scientists" observing homeowners "interact" with their kitchen cabinets. New features today include deep drawers for cookware, pull-out shelves to avoid excess bending, sponge trays on the front of sink cabinets, pull-out hideaway garbage/recycling containers, pull-out spice cabinets, in corner cabinets, vertical storage for cookie sheets, full-extension drawer slides, and drawers and doors with so-called soft-close/positive-close mechanisms enabling drawers to shut quietly, or which shut fully after being pushed only partially.

## 2. RAW MATERIALS

To make a modular kitchen, the raw materials used are:

- Wooden Laminates and Veneers
- Timber
- Medium Density Fibre board (MDF)
- Plywood
- Glue and paints
- Polishes
- Screws and nails

## 3. MANUFACTURING PROCESS

### 3.1. Framing

Cabinets may be either face-frame or frameless in construction.

a. **Face frame cabinets.** Traditional cabinets are constructed using face frames which typically consist of narrow strips of hardwood framing the cabinet box opening. Cabinet carcasses were traditionally constructed with a separate face frame until the introduction of modern engineered wood such as particle board and Medium Density Fibre board along with glues, hinges and fasteners required to join them. A face frame ensures squareness of the cabinet front. It also increases rigidity and provides a mounting point for hinges.

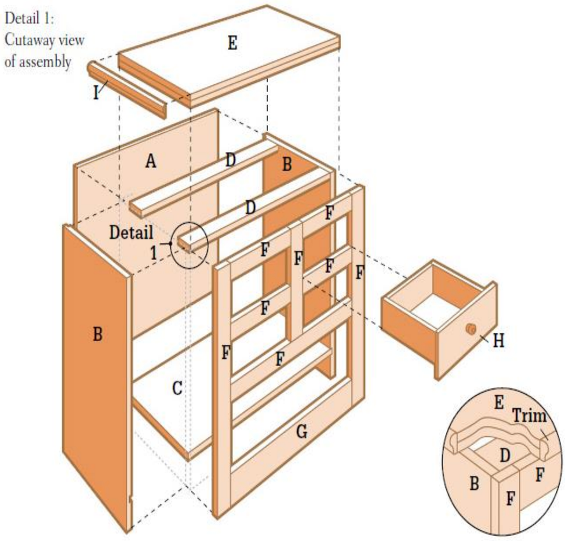


**Figure.1:** Face frame cabinet construction

b. **Frameless cabinets:** Frameless (“full-access") cabinets utilize the carcase side, top, and bottom panels to serve same functions as do face-frames in traditional cabinets. In general, frameless cabinets provide better utilization of space than face-frame cabinets. Frameless cabinets rely on manufacturing methods that permit the production of modern cabinet hardware (hinges and slides) and engineered wood products (for strength, dimensional tolerance, and stability). The intent of the frameless design is to achieve a more streamlined appearance and a more efficient use of space, with ergonomically designed moving components such as drawers, trays, and pull-out cabinets providing better access to interior components.

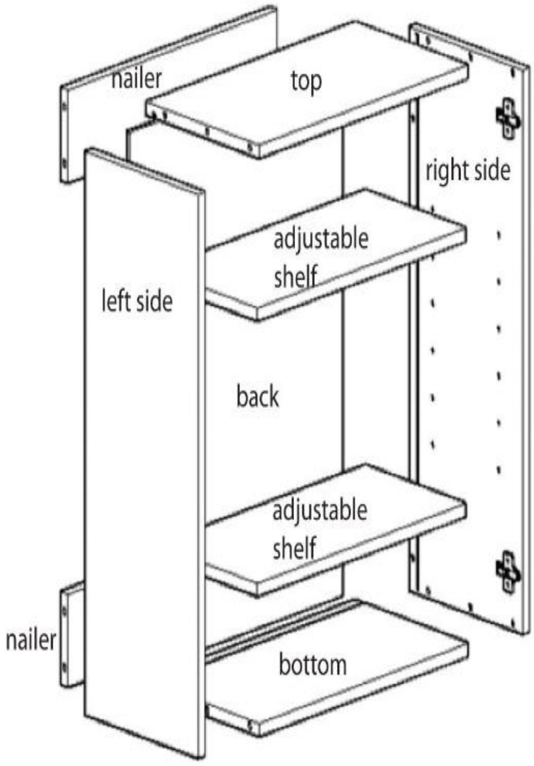


**Figure. 2:** Frameless cabinet construction

### 3.2. Cabinet Detailing

a. **Door Mounting:** For both face-frame and frameless kitchen cabinets, it is conventional for cabinet doors to overlay the cabinet carcass. Face-frame cabinets allow for various door mounting options. The two types (frameless and full-overlay face-frame cabinets) have a similar installed appearance (when doors are closed), both may use European cup hinges, and both tend to use decorative door and drawer pulls.

b. **Material options:** Frameless cabinets, which exhibit a modern appearance in keeping with the design movement of "minimalism," are typically constructed of particle board, which features a high degree of dimensional stability, adherence to dimensional standards, absence of warping, uniformity, and a lower cost than solid wood and is very strong. Plywood and/or solid wood can also be used in frameless cabinet construction, generally at a higher cost.

### 3.3. Drawers and Trays

A functional design objective for cabinet interiors involves maximization of useful space and utility in the context of the kitchen workflow. Drawers and trays in lower cabinets permit access from above and avoid uncomfortable or painful crouching.

In face-frame construction, a drawer or tray must clear the face-frame stile and is 2 inches (51 mm) narrower than the available cabinet interior space. The loss of 2 inches is particularly noticeable and significant for kitchens including multiple narrow [15-inch (380 mm) or less] cabinets.

In frameless construction, drawer boxes may be sized nearly to the interior opening of the cabinet providing better use of the available space.

However, the same is not true for trays. Even in the case of frameless construction doors and their hinges when open, block a portion of the interior cabinet width. Since trays are mounted behind the door, trays are typically significantly narrower than drawers.

### 3.4. Cabinet Hardware

Hardware is the term used for metal fittings incorporated into a cabinet extraneous of the wood or engineered wood substitute and the countertop. The most basic hardware consists of hinges and drawer/door pulls, although only hinges are an absolute necessity for a cabinet since pulls can be fashioned of wood or plastic, and drawer slides were traditionally fashioned of wood.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The dimensions should be as per design requirements
- The surfaces should be well polished
- The joints should be tight
- Hinges of doors and drawers should be smooth

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | **a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations<br>**b.** European Union Timber Regulation (EUTR)<br>**c.** General Product Safety Directive (GPSD) | **a.** European Commission<br>**b.** European Chemicals Agency (ECHA) | **a.** Legality of timber<br>**b.** Traceability<br>**c.** Procedural framework<br>**d.** Marking and labelling |
| 2 | United States of America | **a.** The Lacey Act,2008<br>**b.** California Proposition 65, Act of 1986<br>**c.** Bureau of Consumer Protection Regulations<br>**d.** Environment Protection Act (EPA)<br>**e.** Consumer Product Safety Improvement Act (CPSIA), 2008 | **a.** Consumer Product Safety Commission (CPSC)<br>**b.** Animal and Plant Health Inspection Service (APHIS)<br>**c.** United States Environment Protection Agency (EPA)<br>**d.** California State Government<br>**e.** Federal Trade Commission | **a.** Nature of Timber (endangered or not)<br>**b.** The genus and species of tree<br>**c.** Quantity of regulated material<br>**d.** Country of origin<br>**e.** Formaldehyde in wood<br>**f.** Labelling<br>**g.** Children product certificate testing for children furniture |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 3 | Australia | a. Bio Security Import Conditions System (BICON)<br>b. Australian Competition and Consumer Commission Act, 2010<br>c. Illegal Logging Prohibition Act, 2012 | a. Australian Competition and Consumer Commission (ACCC)<br>b. Department of Agriculture, Water and Environment | a. Legality of Logs<br>b. Quality of product<br>c. Pest free product |
| 4 | Middle East | a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations<br>b. SASO Certificate of Conformity | a. Safety, Saudi Standards, Metrology and Quality Organization (SASO) | a. General product safety regulation<br>b. Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **General (Refer Table 1)** | **IS 151: 2017** | Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | European Union | **ISO 4618: 2014** | Paints and varnishes — Terms and definitions |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** | Paints and related materials | Australia | **AS 4491:1997** | Glossary of terms related to timber |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** | Timber technology and utilization of wood, bamboo and cane - Glossary of terms | Australia | **AS 2310** | Glossary of paints and painting terms |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** | Fire safety | Middle East | **ISO 24294: 2021** | Vocabulary for timber |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** | Classification of commercial timbers and their zonal distribution | Middle East | **GSO 450: 1994** | Paints and varnishes |
| **Raw Materials (Refer Table 2.a & 2.b)** | **IS 3087: 2005 (Reaffirmed Year: 2020)** | Particle boards of wood and other lignocelluloses materials (medium density) for general purposes - Specification | European Union | **ISO/TC 218** | Specification for timber |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** | Veneered particle boards - Specification | USA | **ASTM - D4761** | Test method for wood based material |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** | Specification of animal glue | European Union | **ISO/DIS 9665** | Adhesives — Animal glues |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** | Plywood for general purposes | USA | **ASTM 3618: 2005: R2015** | Paint: Detection of lead |
| | **IS 12406: 2021** | Medium density fibre boards for general purpose - Specification | Middle East | **ISO 1096: 2021** | Specification for plywood classification |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** | Indian timbers for furniture and cabinets- Classification | USA | **ASTM 4442- 20** | Moisture content in timber |
| | **IS 6760: 1972 (Reaffirmed Year: 2021)** | Screws | Australia | **AS 2796** | Seasoning of timber |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** | Evaluation of quality of timber | Australia | **AS 1080.2.4 - 1981** | Moisture content in timber |
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** | Code of practice for seasoning of timber | USA | **ASTM D143** | Test methods for timber |
| | - | - | Middle East | **GSO 1424: 2002** | Methods of testing timber |
| **Manufacturing/ Processing (Refer Table 3)** | **IS 3478: 1966 (Reaffirmed Year: 2018)** | Specification for high density wood particle boards | - | - | - |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** | Preservation of timber | - | - | - |
| **Finishing (Refer Table 4.a & 4.b)** | **IS 287: 1993 (Reaffirmed Year: 2017)** | Permissible moisture content for timber used for different purposes | European Union | **ISO 3350: 1975** | Hardness |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** | Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | European Union | **ISO 21887: 2007** | Durability: Use classes |
| | **IS 4116: 1988 (Reaffirmed Year: 2017)** | Wooden cupboards | Middle East | **GSO ISO 21887: 2007** | Durability: Use classes |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** | Code of practice for finishing of wood and wood-based materials, Part 1: Operations and workmanship | Middle East | **GSO CEN/TS 16818** | Durability: Moisture dynamics |
| | - | - | European Union | **ISO 4211 -2: 2013** | Test for surface finishes. |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| Stage | Code | Aspects Covered | Region/ Country | Standard | Aspects Covered |
| | - | - | European Union | **ISO 7170: 2021** | Furniture — Storage units — Determination of strength and durability |

### 6.1. Table 1: (Domestic & International General Standards)

| Stage | Code | Scope |
|---|---|---|
| General | **IS 151: 2017** - Ready mixed paint, spraying, finishing, stoving, enamel for general purposes, colour as required - Specification (second revision) | ▪ Requirements and the methods of sampling and test for ready mixed paint, spraying, finishing, stoving, enamel, for general purposes, colour as required.<br>▪ The material is used for the protection and decoration of metal parts of apparatus, equipment, machines, etc., and is normally applied as a painting system over the appropriate priming and undercoating paints. |
| | **IS 707: 2007 (Reaffirmed Year: 2021)** - Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms | ▪ This standard covers definitions of common terms applicable to timber technology and forest products utilization.<br>▪ NOTE - Botanical features and other purely scientific terms are excluded from the scope of this standard |
| | **IS 6329: 2000 (Reaffirmed Year: 2021)** - Practice for Fire safety | ▪ Covers the fire safety requirements of sawmills, furniture factories, coach and body building works, upholsteries and other wood working workshops, where various kinds of wood working operations are carried out either as a separate trade or as ancillary to any particular industry.<br>▪ Covers fire safety requirements of factories making various varieties of wood products, namely, plywood, hardboards, wood wool, insulation boards, wood flour, etc. |
| | **IS 399: 1963 (Reaffirmed Year: 2020)** - Classification of Commercial Timbers and Their zonal Distribution | ▪ This standard details the zonal distribution of common commercial timbers of India, classified according to their various uses, and gives information on the availability of these timbers and on some of their important properties. |
| | **IS 1303: 1983 (Reaffirmed Year: 2017)** - Paints and related materials | ▪ This standard defines technical terms widely used in paint industry, and includes terms for paints, varnishes, enamels and allied products. |
| | **ISO 4618: 2014** - Paint | ▪ Details of paints which can be used on wooden surfaces |
| | **AS 2310** - Glossary of paints and painting terms | ▪ Glossary of paints |
| | **GSO 450: 1994** - Primary paints | ▪ Vocabulary of all types of primary paints |
| | **ISO 24294: 2021** - Vocabulary for timber | ▪ Details of various types of timber |
| | **AS 4491:1997** - Glossary of terms related to Timber | ▪ Glossary of terms in timber-related Standards |

### 6.2. Table 2.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Materials | **IS 3087: 2005 (Reaffirmed Year: 2020)** - Particle boards of wood and other lignocellulose materials (medium density) for general purposes -Specification | ▪ Covers the requirements of medium density particle boards made of wood and/or other lignocelluloses materials for general purposes, having specific gravity in the range 0.5 to 0.9.<br>▪ Does not cover veneered particleboards, moulded particle boards, high and low density particle boards or particle boards faced by impregnated paper surfaces. |
| | **IS 3097: 2006 (Reaffirmed Year: 2022)** - Veneered particle boards -specification | ▪ Covers the requirements such as grades and types, material, manufacture, dimensions and tests for veneered particle boards. |
| | **IS 852: 1994 (Reaffirmed Year: 2020)** - Specification of glue | ▪ This standard covers requirements of animal glue for general wood-working purposes. |
| | **IS 303: 1989 (Reaffirmed Year: 2018)** -Plywood for general purposes | ▪ This standard covers the requirements of different grades and types of plywood used for general purposes. |
| | **IS 12406: 2021** - Medium density fibre boards for general purpose | ▪ This standard covers the requirements of medium density fibre boards for general purposes having density in the range of600-900 kg/m3.<br>▪ This standard does not cover veneered or laminated or pre-laminated or other specially treated boards, moulded boards, etc. |
| | **IS 13622: 1993 (Reaffirmed Year: 2017)** - Indian timbers for furniture and cabinets- Classification | ▪ This standard covers the general classification of Indian Timber species suitable for furniture and cabinets. It also lays down the general requirements of quality, moisture content and preservative treatment for the timber. |
| | **IS 6760: 1972 (Reaffirmed Year: 2021)** - Slotted countersunk head wood screws | ▪ Requirement and specifications of screws used in wooden furniture |
| | **IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber | ▪ This standard covers methods of conducting following operations(tests) for evaluating working quality of timber**:**<br>a. Planning<br>b. Sanding<br>c. Turning<br>d. Shaping<br>e. Mortising<br>f. Boring<br>▪ The common wood working operations used in the manufacture of timber stores and products. |

| Stage | Code | Scope |
|---|---|---|
| | **IS 1141: 1993 (Reaffirmed Year: 2020)** - Seasoning of timber | ▪ This standard covers classification of timbers for seasoning purposes; preliminary treatment and storage; stacking practice; pre-seasoning treatment; seasoning methods; kiln schedules for seasoning different species of timbers; pre and post-treatment seasoning; kiln operation procedure; measures for control of warp; and inspection, transport and storage of seasoned timber. General guidelines are also included for seasoning of bamboo.<br>▪ The design and construction of seasoning kilns, recommended moisture contents for seasoned timber and methods of determination of moisture content are covered in separate Indian Standards, namely, IS 7315: 1974, IS 287: 1993 (Reaffirmed Year: 2017) and IS 11215 :1991 respectively. |

### 6.3. Table 2.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM - D4761** - Test method for wood-based material | ▪ Cover the determination of the mechanical properties of stress-graded lumber and other wood-based structural materials<br>▪ This standard does not purport to address all of the safety concerns, if say, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use. |
| | **ISO/TC 218** - Specification for Timber | ▪ Standardization of round, sawn and processed timber and timber materials in and for use in all applications, including terminology, specifications and test methods. |
| | **ISO/DIS 9665** - Adhesives — Animal glues | ▪ Specification of glues used in wooden furniture |
| | **GSO ISO 1096: 2021** - Specification for Plywood classification | ▪ Provides systems of classification of plywood panels based on general appearance and principal characteristics. |
| | **ASTM 4442- 20** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **AS 2796** - Timber - Seasoned hardwood | ▪ standard covers classification of seasoned timber and summary of surface finishes for seasoned hardwood |
| | **AS 1080.2.4 – 1981** - Methods of testing timber - Moisture content | ▪ Specifies methods to determine the moisture content in timber |
| | **ASTM D143** - Test methods for timber | ▪ Test methods cover the determination of various strength and related properties of wood by testing small clear specimens.<br>▪ Test methods represent procedures for evaluating the different mechanical and physical properties, controlling factors such as specimen size, moisture content, temperature, and rate of loading.<br>▪ The values stated in inch-pound units are to be regarded as the standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. When a weight is prescribed, the basic inch-pound unit of weight (lb.) and the basic SI unit of mass (Kg) are cited. |

| Stage | Code | Scope |
|---|---|---|
| | **GSO 1424: 2002** - Methods of testing timber | ▪ Methods of Testing of Timber Part1: Standard Methods of Testing Small Clear Specimens of Timber |
| | **ASTM 3618: 2005: R2015** - Paint: Detection of lead | ▪ Detection of lead in paints |

### 6.4. Table 3: (Domestic Standards for Processing)

| Stage | Code | Scope |
|---|---|---|
| Processing | **IS 3478: 1966 (Reaffirmed Year: 2018)** - Specification for high density wood particle boards | ▪ Covers the requirements of high-density wood particle boards in flat sheet or moulded forms. |
| | **IS 401: 2001 (Reaffirmed Year: 2021)** - Preservation of timber | ▪ Covers types of preservatives, their brief description, methods of treatment, choice of treatment for different species of timber for a number of uses and determination of degree of penetration. Prophylactic treatment of logs and sawn timber during storage is also covered.<br>▪ This standard includes only such preservatives and methods of treatment, which have given satisfactory results under the Indian conditions of service. |

### 6.5. Table 4.a: (Domestic Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **IS 287: 1993 (Reaffirmed Year: 2017)** - Permissible moisture content for timber used for different purposes - Recommendations | ▪ This standard covers the recommendations for permissible limits of moisture content based on optimum moisture contents indicated by experimental data and the tolerances permissible in these limits, for different stores in each of the four climatic zones into which the country has been broadly divided for this purpose, with a view to ensuring unimpaired utility of the stores as a result of climatic changes. |
| | **IS 4116: 1988 (Reaffirmed Year: 2017)** - Wooden shelving cabinets (adjustable type) -Specification | ▪ Requirements for materials, sizes, construction and dimensions of wooden cabinets with doors. |
| | **IS 2338: Part 1: 1967 (Reaffirmed Year: 2020)** - Code of practice for finishing of wood and wood- based materials, Part 1: Operations and workmanship | ▪ Part 1 Operations and workmanship<br>▪ Part 2 Schedules |
| | **IS 5807: Part 1: 1975 (Reaffirmed Year: 2017)** - Methods of test for clear finishes for wooden furniture, Part 1: Resistance to dry heat | ▪ Methods of test for clear finishes for wooden furniture: Part 1 Resistance to dry heat |

### 6.6. Table 4. b: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ISO 3350: 1975** - Wood - Determination of static hardness | ▪ Specifies a method for the determination of the static hardness of wood. |
| | **ISO 21887: 2007** - Durability of wood and wood-based products - Use classes | ▪ Methods to analyse the durability of the wood products |

| Stage | Code | Scope |
|---|---|---|
| | **GSO ISO 21887: 2007** - Durability of wood and wood-based products - Use classes | ▪ Methods to analyse the durability of the wood products |
| | **GSO CEN/TS 16818** - Durability of wood and wood-based products - Moisture dynamics of wood and wood-based products | ▪ Specifies methods to determine the moisture content in wood and wood-based materials. |
| | **ISO 4211 -2: 2013** - Test for surface finishes | ▪ Specifies a method for the assessment of the resistance to wet heat of all rigid furniture surfaces regardless of materials.<br>▪ The test is intended to be carried out on a part of the finished furniture, but can be carried out on test panels of the same material, finished in an identical manner to the finished product and of a size sufficient to meet the requirements of the test. |
| | **ISO 7170: 2021** - Furniture | ▪ This International Standard specifies test methods for determining the strength and durability of storage units that are fully assembled and ready for use, including their movable and non-movable parts.<br>▪ The tests consist of the application, to various parts of the unit, of loads, forces and velocities simulating normal functional use, as well as misuse, that might reasonably be expected to occur.<br>▪ These results may be used to represent the performance of production models provided that the tested model is representative of the production model. |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES

1. Product information- http://odopup.in/en/article/Raebareli
2. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
3. ISO Standards: https://www.iso.org/obp/ui#search
4. EU : https://www.en-standard.eu/
5. USA : https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
6. AUS : https://www.standards.org.au/
7. GULF: https://www.gso.org.sa/store/
8. NABL accredited laboratories : https://nabl-india.org/
9. Domestic Legislation/Regulation : https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
10. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/environment/forests/timber_regulation.htm
    b. USA : https://www.compliancegate.com/united-states-wood-and-bamboo-product-regulations/
    c. AUS: https://www.agriculture.gov.au/import/goods/timber

## METAL INLAY, SAHARANPUR

### 1. INTRODUCTION

Metal inlay work is a unique art form used for decorating wooden boxes and other wooden items. Inlay covers a wide range of techniques in sculpture and the decorative arts for inserting pieces of contrasting, often coloured materials into depressions in a base object to form ornaments or pictures that normally are flush with the wooden surface.

The wood crafts of Saharanpur are renowned the world over and have been given a GI tag, which further enhances their presence in the global markets as a unique product.

The wood used in making these products are available locally and are used with great skill by the craftsmen to create a vast and amazing array of artistic handicrafts such as jewellery boxes, pen cases, snuff boxes etc.

### 2. RAW MATERIALS

- **Inlay work:** Marble & Metals like copper, brass or silver wire
- **Wood:** Sheesham, Rose wood, Cedar, Pine, Rubber wood, red sandal and Jackfruit wood
- Mixture of wax and charcoal to fill fine details
- Adhesive

### 3. PRODUCTION PROCESS (WOOD INLAY PROCESS)

a. **Preparing:** The cured and seasoned wood is chosen and roughly cut to the sizes required. The artisans craft the wood into boxes or other decorative items.

b. **Designing:** The artisans trace the patterns to be cut out onto the wood using templates.

c. **Gouging:** Using chisels and other woodworking tools, the craftsmen remove wood pieces in the designated areas. It's in these cavities carefully carved metal pieces or marble chips are inserted. The inlay material determines the depth of the carved out parts.

**Figure. 1:** Metal installed on edges

d. **Cutting the inlay:** Sheet metal or the inlay material such as marble or semi-precious stones are cut into shapes as desired. Using adhesives these inlay materials are installed into the carved cavities.

e. **Decorating the edges:** Depending on the designs, the artisans reinforce the edges of the boxes with metals.

f. **Finishing:** A final lacquer coating is given to the wooden surface to give it a glossy finish.

## 4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The wood chosen should not have any knots and free from decays, cracks etc.
- The inlay material should be clean, smooth, should have no joints in the wire and free from surface defects
- Polish and varnish applied should be uniform, devoid of patches

## 5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

### 5.1. Domestic

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | India | NA | NA | NA |

### 5.2. 5.2 International

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| 1 | European Union | a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) | a. European Chemicals Agency (ECHA)<br>b. European Commission | a. Legality of timber<br>b. Traceability<br>c. Marking and labelling<br>d. Hazardous metals |

| S. No. | Country | Regulation | Regulatory Body | Remarks |
|---|---|---|---|---|
| | | Regulation (EC) 1907/2006<br>**b.** European Union Timber Regulation (EUTR)<br>**c.** General Product Safety Directive(GPSD)<br>**d.** Restriction of Hazardous Substances (RoHS) Regulation (EC) 2002/95 | | |
| 2 | United States of America | **a.** Animal and Plant Health Inspection Service (APHIS)<br>**b.** California Proposition 65, Act 0f 1986<br>**c.** Restriction of Hazardous Substances(RoHS) Regulations | **a.** Consumer Product Safety Commission(CPSC)<br>**b.** California State | **a.** Nature of Timber (endangered or not)<br>**b.** The genus and species of tree<br>**c.** Quantity of regulated material |
| 3 | Australia | **a.** Bio security Import Conditions system (BICON)<br>**b.** Australian Competition and Consumer Commission Act 2010 | **a.** Australian Competition and Consumer Commission (ACCC) | **a.** Legality of logs<br>**b.** Quality of product<br>**c.** Pest free product |
| 4 | Middle East | **a.** SASO Certificate of Conformity<br>**b.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations | **a.** Saudi Standards, Metrology and Quality Organization (SASO) | **a.** General product safety regulation<br>**b.** Labelling Requirement |

## 6. VOLUNTARY STANDARDS

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| **Raw Materials (Refer Table 1.a & 1.b)** | **IS 12994: 1990 (Reaffirmed Year: 2020)** | Adhesive glue | USA | **ASTM D6880_D6880 M-19** | Standard specification for wood boxes |
| | **IS 410: 1977 (Reaffirmed Year: 2021)** | Cold rolled brass sheet, strip and foil | USA | **ASTM B0036_B0036 M-18** | Standard specification for brass plate, sheet, strip and rolled bar |
| | **IS 2283: 2000 (Reaffirmed Year: 2021)** | Nickel silver sheet, strip specification | European Union | **PD CR 14376:2002** | Adhesives for paper and board |

| Domestic Standards | | | International Standards | | |
|---|---|---|---|---|---|
| **Stage** | **Code** | **Aspects Covered** | **Region/ Country** | **Standard** | **Aspects Covered** |
| | **IS 3052: 1986 (Reaffirmed Year: 2021)** | Dimensions and tolerances for wrought copper and copper alloy sheet | European union | **ISO 15605: 2000** | Methods for sampling adhesives and related products |
| | **IS 2112: 2014 (Reaffirmed Year: 2019)** | Silver and silver alloys, jewellery/artefacts - Fineness and marking | Australia | **AS/NZS 2754.1: 2016** | Adhesives for timber and timber products |
| | **IS 2279: 1980(Reaffirmed Year: 2019)** | Fine silver bar, sheet, wire, granules and token | Middle East | **GSO ISO 1190-1:2013** | Copper and Copper Alloys - Code Of designation |
| | **IS 7814: 2005 (Reaffirmed Year: 2021)** | Phosphor bronze sheet, strip and Foil - Specificat8ion | - | - | - |
| | **IS 12289: 1988 (Reaffirmed Year: 2019)** | Hand held engraving tools | - | - | - |
| **Finishing (Refer Table 2)** | - | - | USA | **ASTM D333** | Guide for clear and pigmented lacquers |
| | - | - | Middle East | **GSO ASTM D3924: 2011** | Specification for standard environment for conditioning and testing paint, varnish, lacquer and related materials |

### 6.1. Table 1.a: (Domestic Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| **Raw Materials** | **IS 12994: 1990 (Reaffirmed Year: 2020)** -Requirements and methods of sampling and test for liquid and paste type epoxy adhesives | ▪ Requirements and methods of sampling and test for liquid and paste type epoxy adhesives for performance: a) up to 50°C, and (b) up to 100°C<br>▪ Referenced documents |
| | **IS 410: 1977 (Reaffirmed Year: 2021)** - Requirements of three alloys of cold rolled brass sheet, strip and foil required for engineering and general purposes | ▪ Requirements of three alloys of cold rolled brass sheet, strip and foil required for engineering and general purposes and designated as CuZn30, CuZn37 and CuZn40<br>▪ Adjunct standards and referenced documents |
| | **IS 2283: 2000 (Reaffirmed Year: 2021)** - Requirements for Nickel Silver sheet, strip and foil for general purposes | ▪ Requirements for Nickel Silver sheet, strip and foil for general purposes<br>▪ Referenced documents |
| | **IS 3052: 1986 (Reaffirmed Year: 2021)** - Standard lays down the dimensions and tolerances for wrought copper and copper alloys in the form of sheet, strip and foil for general engineering purposes. | ▪ Dimensions and tolerances for wrought copper and copper alloys in the form of sheet, strip and foil for general engineering purposes<br>▪ Referenced documents |
| | **IS 2112: 2014 (Reaffirmed Year: 2019)** - Specifies three grades of silver,<br>silver used in manufacture of jewellery/artefacts | ▪ Specification for three grades of fine silver,<br>▪ Six grades of silver alloys used in the manufacture of jewellery/artefacts of silver based on their silver content<br>▪ Referred standards |

| Stage | Code | Scope |
|---|---|---|
| | **IS 2279: 1980(Reaffirmed Year: 2019)** - Standard covers the requirements of fine silver in the form of bar, sheet, wire, granules and token. It, however, does not cover wires for electrical contacts. | ▪ Requirements of fine silver in the form of bar, sheet, wire, granules and token<br>▪ Exclusions<br>▪ Referred documents |
| | **IS 7814: 2005 (Reaffirmed Year: 2021)** - Standard covers the chemical composition, mechanical properties and other requirements for four grades of rolled phosphor bronze sheet, strip and foils. | ▪ Requirement for four grades of rolled phosphor bronze sheet, strip and foils.<br>▪ Referenced documents |
| | **IS 12289: 1988 (Reaffirmed Year: 2019)** - Specification for hand held electric engraving tools | ▪ Specification of hand held electric engraving tools. |

### 6.2. Table 1.b: (International Standards for Raw Materials)

| Stage | Code | Scope |
|---|---|---|
| Raw Material | **ASTM D6880_D6880M-19** - Specification covers the fabrication methods for the production of wood boxes | ▪ Specification covers the fabrication of wood boxes<br>▪ Units of measurement |
| | **ASTM B0036_B0036M-18** - This specification establishes the requirements for brass plate, sheet, strip, and rolled bar | ▪ Specification establishes the requirements for brass plate, sheet, strip, and rolled bar of the following alloys<br>▪ Referenced documents |
| | **ISO 15605: 2000** - Methods for sampling adhesives and related products, in order to obtain uniform samples of convenient size which are adequately representative of the product being sampled | ▪ Specifies methods for sampling adhesives and related products<br>▪ Normative references<br>▪ Terms |
| | **AS/NZS 2754.1: 2016** - Requirements for adhesives suitable for use in the plywood and laminated veneer lumber | ▪ Adhesives for Timber and timber products |

### 6.3. Table 2: (International Standards for Finishing)

| Stage | Code | Scope |
|---|---|---|
| Finishing | **ASTM D333** - Test methods cover procedures for testing lacquers and lacquer coatings | ▪ Procedures for testing lacquers and lacquer coatings.<br>▪ Units of measurement |
| | **GSO ASTM D3924: 2011:** This specification defines the standard atmospheres for normal conditioning and testing of paint, varnish, lacquer, and related materials, at approximately ambient conditions | ▪ Standard atmospheres for normal conditioning and testing of paint, varnish, lacquer, and related materials, at approximately ambient conditions<br>▪ Referenced documents |

**Note:** The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

## REFERENCES


1. Product information: http://odopup.in/en/article/saharanpur
2. Product information: https://en.wikipedia.org/wiki/Inlay
3. Indian Standards under Compulsory Certification: https://www.bsbedge.com/mandatory-indian-standards
4. Domestic Standards for Wooden engraving and raw materials: https://standardsbis.bsbedge.com/
5. BIS product conformation overview: https://bis.gov.in/index.php/product-certification/product-certification-overview/
6. Mandatory BIS standards: https://www.bsbedge.com/mandatory-indian-standards
7. ISO Standards: https://www.iso.org/obp/ui#search
8. International Standards:
    a. EU: https://www.en-standard.eu/
    b. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
    c. AUS: https://www.standards.org.au/
    d. GULF: https://www.gso.org.sa/store/
9. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

## APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

### 1. PRODUCT TESTING

#### What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

#### Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

#### Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

**Where can you get the list of accredited/approved laboratories?**

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

## 2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

**a. Management Systems Certification**

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

**b. Product Certification**

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

### Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

### Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

**Where can you get the list of accredited/approved laboratories?**

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

**National Accreditation Board for Certification Bodies (NABCB):** http://nabcb.qci.org.in/

# APPENDIX II: IMPORTANT INSTITUTIONS

| WOODEN PRODUCTS | | | | |
|---|---|---|---|---|
| **S. No.** | **Institute** | **Areas of Expertise** | **Postal Address** | **Contact** |
| 1 | Institute of Wood Science & Technology | ▪ Material Testing<br>▪ Consultancy | Institute of Wood Science & Technology, 18th Cross, Malleshwaram, Bengaluru, Karnataka 560003, India | Email: dir_iwst@icfre.org<br>Phone: 22190101<br>Website: https://iwst.icfre.gov.in/ |
| 2 | Forest Research Institute | ▪ Forest Conservation<br>▪ Production and Supply of timber<br>▪ Planting Stock improvement<br>▪ Forestry Education<br>▪ Technology for Eco Friendly Preservatives | Forest Research Institute, Chakarata Rd, New Forest, P.O, Indian Military Academy, Dehradun, Uttarakhand 248006 | Email: singhkp@icfre.org<br>Phone: 0135-2224315<br>Website: https://fri.icfre.gov.in/ |
| 3 | Indian Plywood Industries Research and Training Institute (IPIRTI) | ▪ Consultancy<br>▪ Training & Workshops<br>▪ Technology Transfer<br>▪ Testing | Indian Plywood Industries Research & Training Institute (Autonomous body of Ministry of Environment, Forest & Climate Change, Government of India) Post Bag No.2273, Tumkur Road, Yeshwanthpur PO, Bangalore - 560 022<br>Phone: 080 28394231,32,33, 30534000<br>Fax: 080 28396361 | Email: contactus@ipirti.gov.in<br>Phone: 080-28394321<br>Website: https://ipirti.gov.in/ |

ONE DISTRICT ONE PRODUCT

UTTAR PRADESH

ODOP Cell, Directorate of Industries, Niryat Bhawan, First Floor, 8 Cantt Road, Qaiserbagh, Lucknow – 226001